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389. On a Locus derived from Two Conics.

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I suppose in particular that the two conics are

$$x^2 + my^2 - 1 = 0, \quad mx^2 + y^2 - 1 = 0, \tag{1}$$

the equation of the quartic is

$$4(\lambda+1)^2 m^2 (x^2 + my^2 - 1)(mx^2 + y^2 - 1) - \{(m^2+m)(x^2+y^2) - m^2 - 1\}^2 = 0, \tag{2}$$

or putting $\lambda = \dfrac{k}{m^2+1}$, this is

$$(\lambda+1)^2\{x^2+y^2-a\}^2 - (x^2+my^2-1)(mx^2+y^2-1) = 0. \tag{3}$$

To fix the ideas, suppose that m is positive and > 1, so that each of the conics is an ellipse, the major semi-axis being $= 1$, and the minor semi-axis being $= \dfrac{1}{\sqrt{m}}$.

For any real value of k the coefficient λ is positive, and it may accordingly be assumed that λ is positive.

We have $\dfrac{1}{m(m+1)} < a < 1$, or the radius of the circle is intermediate between the semi-axes of the ellipses, hence the points of contact on each ellipse are real points.

Writing for shortness $a = \dfrac{m^2+1}{\lambda(m+1)}$, the equation is

$$(x^2+my^2-1)(mx^2+y^2-1) - \lambda(x^2+y^2-a)^2 = 0. \tag{4}$$

For the points on the axis of x, we have

$$(x^2-1)(mx^2-1) - \lambda(x^2-a)^2 = 0, \tag{5}$$

that is

$$(m-\lambda)x^4 + \{-(1+m) + 2\lambda a\}x^2 + (1-\lambda a^2) = 0, \tag{6}$$

and thence

$$(m-\lambda)x^2 = \tfrac{1}{2}(1+m) - \lambda a \pm \tfrac{1}{2}\sqrt{(m-1)^2 + 4\lambda(1-a)(1-ma)}, \tag{7}$$

or, substituting for a its value, this is

$$(m-\lambda)x^2 = \tfrac{1}{2}(m+1) - \frac{\lambda(m^2+1)}{\lambda(m+1)} \pm \tfrac{1}{2}\sqrt{(m-1)^2 + 4\lambda(1-a)(1-ma)}. \tag{8}$$

Remarking that the values of λ: 0, $\dfrac{1}{(m+1)^2}$, $\dfrac{m}{(m+1)^2}$, m, $\dfrac{1}{4}(m+1)^2$ are in the order of increasing magnitude.

and considering successive values of λ; first the value $\lambda = \dfrac{1}{(m+1)^2}$, we have

$$(m-\lambda)x^2 = \tfrac{1}{2}(m+1) - \frac{1}{m+1} \pm \tfrac{1}{4}\sqrt{(m-1)(m-\tfrac{1}{m})}, \tag{9}$$

or observing that

$$(m+1)^2(m+\tfrac{1}{m}-2) = (m+1)^2 - (m-1)^2 = -(m-1)(m^2-1) = (m-1)(m-\tfrac{1}{m}), \tag{10}$$

this is

$$(m-\lambda)x^2 = 0, \tag{11}$$

or

$$x^2 = \frac{(m-1)(m-\frac{1}{m})}{(m+1)^2} \cdot \frac{(m+1)^2}{m(m+1)} = 0, \tag{12}$$

or, what is the same thing,

$$x^2 - \frac{(m-1)(m^3+2m^2-1)}{m(m+1)^2} = 0, \quad y^2 - \frac{(m-1)(m^3+2m^2-1)}{m(m+1)^2} = 0. \tag{13}$$

The next critical value is $\lambda = m$.
The curve here is

$$(x^2+my^2-1)(mx^2+y^2-1) - m(x^2+y^2-a)^2 = 0, \tag{14}$$

that is

$$m(x^2+y^2)^2 + (1+m^2)x^2y^2 - (m+1)(x^2+y^2) + 1 \\ - m(x^2+y^2)^2 + 2ma(x^2+y^2) - ma^2 = 0, \tag{15}$$

that is

$$(m-1)^2x^2y^2 + (2ma-m-1)(x^2+y^2) + 1 - ma^2 = 0, \tag{16}$$

or, substituting for a its value,

$$2ma - m - 1 = \frac{2m}{m+1} - (m+1) = \frac{-(m-1)^2}{m+1}, \tag{17}$$

$$1-ma^2 = 1-\frac{m(m^2+1)^2}{m^2(m+1)^2} = \frac{(m+1)^2(m^2+1) - m(m^2+1)^2}{m(m+1)^2} = \frac{(m^2+1)(-m^3+m^2+2m+1)}{m(m+1)^2}, \tag{18}$$

$$= -\frac{(m^2+1)(m^3-m^2-2m-1)}{m(m+1)^2}, \tag{19}$$

the equation is

$$(m-1)^2x^2y^2 - \frac{(m-1)^2}{m+1}(x^2+y^2) - \frac{(m^2+1)(m^3-m^2-2m-1)}{m(m+1)^2} = 0, \tag{20}$$

or, as this may also be written,

$$x^2y^2 - \frac{1}{m+1}(x^2+y^2) - \frac{(m^2+1)(m^3-m^2-2m-1)}{m(m-1)^2(m+1)^2} = 0. \tag{21}$$

which has a pair of imaginary asymptotes parallel to the axis of x, and a like pair parallel to the axis of y, or what is the same thing, the curve has two isolated points at infinity, one on each axis.

The next critical value is $\lambda = \frac{1}{4}(m+1)^2$; the curve here reduces itself to the four lines

$$(x+y)^2 = \frac{m+1}{m}, \quad (x-y)^2 = \frac{1}{m}, \tag{21}$$

and it is to be observed that when λ exceeds this value, or say $\lambda > \frac{1}{4}(m+1)^2$, the curve has no real point on either axis; but when $\lambda = \infty$, the curve reduces itself to $(x^2+y^2-a)^2 = 0$, i.e. to the circle $x^2+y^2 = a$ twice repeated, having in this special case real points on the two axes.

It is now easy to trace the curve for the different values of λ.

The curve lies in every case within the unshaded regions of the figure (except in the limiting cases after-mentioned); and it also touches the two ellipses and the four lines at the eight points k, at which points it also cuts the circle; but it does not cut or touch the four lines, the two ellipses, or the circle, except at the points k.

Considering λ as varying by successive steps from 0 to ∞:

$\lambda = 0$, the curve is the two ellipses.

$\lambda < \dfrac{1}{(m+1)^2}$, the curve consists of two ovals, an exterior sinuous oval lying in the four regions a and the four regions b; and an interior oval lying in the region c.

$\lambda = \dfrac{1}{(m+1)^2}$, there is still a sinuous oval as above, but the interior oval has dwindled to a conjugate point at the centre.

$\lambda = \dfrac{m}{(m+1)^2}$, there is still a sinuous oval as above, but the interior oval has dwindled to a conjugate point at the centre. For this critical value, the curve acquires the four lines as bitangents, touching each of them at two consecutive points k.

$\dfrac{m}{(m+1)^2} < \lambda < m$, there is no interior oval, but only a sinuous oval as above; which, as λ increases, approaches continually nearer to the four sides of the square.

$\lambda = m$, there is no change in the general form, but the curve has for this value of λ, two conjugate points, one on each axis at infinity.

$\lambda = \frac{1}{4}(m+1)^2$, the curve becomes the four lines.

$\lambda > \frac{1}{4}(m+1)^2$, the curve lies wholly in the four regions a and the four regions e, consisting thereof of four detached sinuous ovals. As λ deviates less from the value $\frac{1}{4}(m+1)^2$, each oval approaches more nearly to the infinite trilateral formed by the side and infinite line-portions which bound the regions d, e to which the oval belongs. And as λ departs from the limit $\frac{1}{4}(m+1)^2$, and approaches to ∞, each sinuous oval approaches more nearly to the circular arc which separates the two regions d, e, which contains the sinuous oval. Finally, $\lambda = \infty$, the curve is the circle twice repeated.

390. Theorem relating to the Four Conics which touch the same Two Lines and pass through the same Four Points.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. VIII (1867), pp. 162–167.]

390. [Date Nov. 13, 1867.]

I.

The sides of the triangle formed by the given points meet one of the given lines in three points, say P, Q, R; and on this same line we have four points of contact, say A_1, A_2, A_3, A_4; any two pairs, say A_1, A_2; A_3, A_4, form with a properly selected pair, say Q, R, out of the above-mentioned three points, an involution; and we have thus the three involutions

$$\{A_1, A_2;\ A_3, A_4;\ Q, R\},\quad \{A_1, A_2;\ A_3, A_4;\ R, P\},\quad \{A_1, A_2;\ A_3, A_4;\ P, Q\}. \tag{1}$$

To prove this, let $x = 0$, $y = 0$ be the equations of the given lines, and take for the equations of the sides of the triangle formed by the given points

$$bx + ay - ab = 0,\quad b'x + a'y - a'b' = 0,\quad b''x + a''y - a''b'' = 0; \tag{2}$$

the equation of any one of the four conics may be written

$$\frac{Lab}{bx + ay - ab} + \frac{L'a'b'}{b'x + a'y - a'b'} + \frac{L''a''b''}{b''x + a''y - a''b''} = Kxyz, \tag{3}$$

and if this touches the axis of x, say at the point $x = a$, then we must have

$$\frac{La}{x - a} + \frac{L'a'}{x - a'} + \frac{L''a''}{x - a''} = K(x - a). \tag{4}$$

or, assuming as we may do, $K = -(a' - a'')(a'' - a)(a - a')$, this gives

$$La = (a - a)^2(a' - a''), \quad L'a' = (a' - a)^2(a'' - a), \quad L''a'' = (a'' - a)^2(a - a'). \tag{5}$$

But in the same manner, if the conic touch the axis of y, say at the point $y = \beta$, we have

$$Lb = (b - \beta)^2(b' - b''), \quad L'b' = (b' - \beta)^2(b'' - b), \quad L''b'' = (b'' - \beta)^2(b - b'). \tag{6}$$

We have

$$\begin{aligned} &b(a - \alpha)^2(a' - a'') : b'(a' - a)^2(a'' - a) : b''(a'' - a)^2(a - a') \\ &\qquad = a(b - \beta)^2(b' - b'') : a'(b' - \beta)^2(b'' - b) : a''(b'' - \beta)^2(b - b'); \end{aligned} \tag{7}$$

and thence

$$\frac{(a - a)\sqrt{bb'}}{a - a} : \frac{(a' - a)\sqrt{b'b''}}{a' - a} : \frac{(a'' - a)\sqrt{b''b}}{a'' - a} = (b - \beta)(b' - b'') : (b' - \beta)(b'' - b) : (b'' - \beta)(b - b'). \tag{8}$$

Putting

$$P = ab(a' - a'')(b' - b''), \quad P' = a'b'(a'' - a)(b'' - b), \quad P'' = a''b''(a - a')(b - b'), \tag{9}$$

we have

$$(a - a)\sqrt{\tfrac{P}{a}} : (a' - a)\sqrt{\tfrac{P'}{a'}} : (a'' - a)\sqrt{\tfrac{P''}{a''}} = (b - \beta)(b' - b'') : (b' - \beta)(b'' - b) : (b'' - \beta)(b - b'), \tag{10}$$

and thence

$$\frac{(a - a)\sqrt{P}}{a} : \frac{(a' - a)\sqrt{P'}}{a'} : \frac{(a'' - a)\sqrt{P''}}{a''} = (b - \beta)(b' - b'') : (b' - \beta)(b'' - b) : (b'' - \beta)(b - b'). \tag{11}$$

which gives

$$(a - a)\sqrt{P} : (a' - a)\sqrt{P'} : (a'' - a)\sqrt{P''} = \frac{b - \beta}{\sqrt{aa'}}(b' - b'') : \frac{b' - \beta}{\sqrt{a'a''}}(b'' - b) : \frac{b'' - \beta}{\sqrt{a''a}}(b - b'). \tag{12}$$

and we have in like manner

$$(b - \beta)\sqrt{P} : (b' - \beta)\sqrt{P'} : (b'' - \beta)\sqrt{P''} = \frac{a - a}{\sqrt{bb'}}(a' - a'') : \frac{a' - a}{\sqrt{b'b''}}(a'' - a) : \frac{a'' - a}{\sqrt{b''b}}(a - a'); \tag{13}$$

but the first of these equations is alone required for the present purpose.

Putting for shortness

$$P = a^2\lambda, \quad P' = a'^2\lambda', \quad P'' = a''^2\lambda'', \tag{14}$$

the equation is

$$(a - a)\sqrt{(\lambda)} + (a' - a)\sqrt{(\lambda')} + (a'' - a)\sqrt{(\lambda'')} = 0, \tag{15}$$

and by attributing the signs + and − to the radicals, we have, corresponding to the four conics, the equations

$$(a-a_1)\sqrt{(\lambda)}+(a'-a_1)\sqrt{(\lambda')}+(a''-a_1)\sqrt{(\lambda'')}=0, \tag{14}$$
$$-(a-a_2)\sqrt{(\lambda)}+(a'-a_2)\sqrt{(\lambda')}+(a''-a_2)\sqrt{(\lambda'')}=0, \tag{15}$$
$$(a-a_3)\sqrt{(\lambda)}-(a'-a_3)\sqrt{(\lambda')}+(a''-a_3)\sqrt{(\lambda'')}=0, \tag{16}$$
$$(a-a_4)\sqrt{(\lambda)}+(a'-a_4)\sqrt{(\lambda')}-(a''-a_4)\sqrt{(\lambda'')}=0, \tag{17}$$

where a_1, a_2, a_3, a_4 are the values of a for the four conics respectively.

Eliminating a we obtain the system of three equations

$$(2a-a_1-a_2)\sqrt{(\lambda)}+(a_2-a_1)\sqrt{(\lambda')}+(a_2-a_1)\sqrt{(\lambda'')}=0, \tag{18}$$
$$(a_3-a_4)\sqrt{(\lambda)}+(2a'-a_3-a_4)\sqrt{(\lambda')}+(a_3-a_4)\sqrt{(\lambda'')}=0, \tag{19}$$
$$(a_1+a_2-a_3-a_4)\sqrt{(\lambda)}+(a_1+a_3-a_2-a_4)\sqrt{(\lambda')}+(a_1+a_4-a_2-a_3)\sqrt{(\lambda'')}=0, \tag{20}$$

and then eliminating the radicals we have

$$\begin{vmatrix} 2a-a_1-a_2 & a_2-a_1 & a_2-a_1 \\ a_3-a_4 & 2a'-a_3-a_4 & a_3-a_4 \\ a_1+a_2-a_3-a_4 & a_1+a_3-a_2-a_4 & a_1+a_4-a_2-a_3 \end{vmatrix}=0, \tag{21}$$

which is in fact

$$\begin{vmatrix} 1 & a+a' & aa' \\ 1 & a_1+a_4 & a_1a_4 \\ 1 & a_2+a_3 & a_2a_3 \end{vmatrix}=0, \tag{22}$$

as may be verified by actual expansion; the transformation of the determinant is a peculiar one.

The foregoing result was originally obtained as follows, viz. writing for a moment

$$a\sqrt{(\lambda)} + a'\sqrt{(\lambda')} + a''\sqrt{(\lambda'')} = \theta, \quad \sqrt{(\lambda)} + \sqrt{(\lambda')} + \sqrt{(\lambda'')} = \Phi, \tag{20}$$

the four equations are

$$\theta - a_1\Phi = 0, \quad \theta - a_2\Phi = 2(a - a_2)\sqrt{(\lambda)}, \quad \theta - a_3\Phi = 2(a' - a_3)\sqrt{(\lambda')}, \quad \theta - a_4\Phi = 2(a'' - a_4)\sqrt{(\lambda'')}. \tag{21}$$

These give

$$(a_1 - a_2)\theta = 2(a - a_2)\sqrt{(\lambda)}, \tag{22}$$

$$(a_1 - a_3)\theta = 2(a' - a_3)\sqrt{(\lambda')}, \tag{23}$$

$$(a_1 - a_4)\theta = 2(a'' - a_4)\sqrt{(\lambda'')}. \tag{24}$$

From the last equation we have

$$\begin{aligned}(a_1 - a_4)\Phi = 2\{\theta - a_4\sqrt{(\lambda)} - a'\sqrt{(\lambda')}\} - 2a_4\{\Phi - \sqrt{(\lambda)} - \sqrt{(\lambda')}\}\\ = 2(a - a_4)\Phi - 2(a - a_4)\sqrt{(\lambda)} - 2(a' - a_4)\sqrt{(\lambda')};\end{aligned} \tag{25}$$

that is

$$(a_1 - a_4)\Phi - 2(a - a_4)\sqrt{(\lambda)} - 2(a' - a_4)\sqrt{(\lambda')} = 0; \tag{26}$$

or substituting for $\sqrt{(\lambda)}$, $\sqrt{(\lambda')}$ their values in terms of Φ, we find

$$\frac{(a - a_4)(a_1 - a_2)}{a - a_2} + \frac{(a' - a_4)(a_1 - a_3)}{a' - a_3} = a_1 - a_4, \tag{27}$$

which may be written

$$\frac{(a - a_4)(a - a_1)}{a - a_2} + \frac{(a' - a_4)(a' - a_1)}{a' - a_3} = 0; \tag{28}$$

that is

$$(a_2 - a_1)\left(1 + \frac{a_2 - a_1}{a - a_2}\right)(a - a_4) + (a_3 - a_1)\left(1 + \frac{a_3 - a_1}{a' - a_3}\right)(a' - a_4) = 0; \tag{29}$$

or again

$$(a - a_2)(a - a_4) + (a' - a_3)(a' - a_4) + (a_2 - a_1)(a - a_4)\frac{a - a_2}{a - a_2} + (a_3 - a_1)(a' - a_4)\frac{a' - a_3}{a' - a_3} = 0; \tag{30}$$

that is

$$(a - a_2)(a - a_4) + (a' - a_3)(a' - a_4) + (a_2 - a_1)(a - a_4) + (a_3 - a_1)(a' - a_4) = 0; \tag{31}$$

or finally

$$(a_2 - a_1)(a - a_4)(a' - a_3) + (a_3 - a_4)(a - a_1)(a' - a_2) = 0, \tag{32}$$

which is a known form of the relation

$$\begin{vmatrix} 1 & a + a' & aa' \\ 1 & a_1 + a_4 & a_1a_4 \\ 1 & a_2 + a_3 & a_2a_3 \end{vmatrix} = 0, \tag{33}$$

which gives the involution of the quantities a, a'; a_1, a_4; a_2, a_3.

We have in like manner

$$\begin{vmatrix} 1 & a' + a'' & a'a'' \\ 1 & a_1 + a_3 & a_1a_3 \\ 1 & a_2 + a_4 & a_2a_4 \end{vmatrix} = 0, \quad \begin{vmatrix} 1 & a'' + a & a''a \\ 1 & a_1 + a_2 & a_1a_2 \\ 1 & a_3 + a_4 & a_3a_4 \end{vmatrix} = 0, \tag{34}$$

which give the involutions of the systems a', a''; a_1, a_3; a_2, a_4 and a'', a; a_1, a_2; a_3, a_4 respectively.

It may be remarked that the equation of the conic passing through the three points and touching the axis of x in the point $x = a$ is

$$\frac{(a-a)^2(a'-a'')b}{bx+ay-ab} + \frac{(a'-a)^2(a''-a)b'}{b'x+a'y-a'b'} + \frac{(a''-a)^2(a-a')b''}{b''x+a''y-a''b''} = 0, \tag{34}$$

and when this meets the axis of y we have

$$\frac{(a-a)^2(a'-a'')}{y-b} + \frac{(a'-a)^2(a''-a)}{y-b'} + \frac{(a''-a)^2(a-a')}{y-b''} = 0. \tag{35}$$

Hence, if this touches the axis of y in the point $y = \beta$, the left-hand side must be

$$R\frac{(a-a)^2(a'-a'') + (a'-a)^2(a''-a) + (a''-a)^2(a-a')}{(y-b)(y-b')(y-b'')}(y-\beta)^2 \tag{36}$$

and equating the coefficients of $\frac{1}{y}$, we have

$$\begin{aligned} &-(a-a)^2(a'-a'') + (a'-a)^2(a''-a) + (a''-a)^2(a-a') \\ &\qquad = [-(a-a)^2(a'-a'') + (a'-a)^2(a''-a) + (a''-a)^2(a-a')](b+b'+b''-2\beta), \end{aligned} \tag{37}$$

or what is the same thing,

$$\begin{aligned} &\frac{(a-a)^2(a'-a'')}{a} + \frac{(a'-a)^2(a''-a)}{a'} + \frac{(a''-a)^2(a-a')}{a''} \\ &\qquad \times \{-(a-2\beta)(a'-a'') + (a'-2\beta)(a''-a) + (a''-2\beta)(a-a')\} = 0, \end{aligned} \tag{38}$$

which gives β in terms of a, that is β_1, β_2, β_3, β_4 in terms of a_1, a_2, a_3, a_4 respectively.

Cambridge, 30 November, 1863.

391. Solution of a Problem of Elimination.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. VIII (1867), pp. 183–185.]

391. [Date Nov. 13, 1867.]
It is required to eliminate x, y from the equations

$$\left.\begin{array}{ccccc} x^4, & x^3y, & x^2y^2, & xy^3, & y^4 \\ a, & b, & c, & d, & e \\ a', & b', & c', & d', & e' \\ a'', & b'', & c'', & d'', & e'' \end{array}\right. = 0. \tag{1}$$

This system may be written

$$x^4 = \Sigma\lambda a, \quad x^3y = \Sigma\lambda b, \quad x^2y^2 = \Sigma\lambda c, \quad xy^3 = \Sigma\lambda d, \quad y^4 = \Sigma\lambda e; \tag{2}$$

or putting

$$y = kx, \tag{3}$$

we have

$$\Sigma\lambda(a+kb) = 0, \quad \Sigma\lambda(b+kc) = 0, \quad \Sigma\lambda(c+kd) = 0, \quad \Sigma\lambda(d+ke) = 0; \tag{4}$$

or, what is the same thing,

$$\lambda(a+kb)+\lambda'(a'+kb')+\lambda''(a''+kb'')=0, \tag{5}$$
$$\lambda(b+kc)+\lambda'(b'+kc')+\lambda''(b''+kc'')=0, \tag{6}$$
$$\lambda(c+kd)+\lambda'(c'+kd')+\lambda''(c''+kd'')=0, \tag{7}$$
$$\lambda(d+ke)+\lambda'(d'+ke')+\lambda''(d''+ke'')=0; \tag{8}$$

and representing the columns

$$\begin{array}{cc|cc|cc} a & b, & a' & b', & a'' & b'' \\ b & c, & b' & c', & b'' & c'' \\ c & d, & c' & d', & c'' & d'' \\ d & e, & d' & e', & d'' & e'' \end{array} \tag{9}$$

by 1, 2, 3, 4, 5, 6; each equation is of the type

$$\lambda(1+k\cdot 2)+\lambda'(3+k\cdot 4)+\lambda''(5+k\cdot 6)=0. \tag{10}$$

Multiplying the several equations by the minors of 135, each with its proper sign, and adding, the terms independent of k disappear, the equation divides by k, and we find

$$\lambda\cdot 2135+\lambda'\cdot 4135+\lambda''\cdot 6135=0; \tag{11}$$

operating in a similar manner with the minors of 246, the terms in k disappear, and we find

$$\lambda\cdot 1246+\lambda'\cdot 3246+\lambda''\cdot 5246=0; \tag{12}$$

again, operating with the minors of $(146+236+245+k\cdot 246)$, we find

$$\begin{aligned} &\lambda\{1236+1245+k(2146+1246)\} \\ &\qquad +\lambda'\{3146+3245+k(4236+3246)\} \\ &\qquad\qquad +\lambda''\{5146+5236+k(6245+5246)\}=0, \end{aligned} \tag{13}$$

where the terms in k disappear, and this is

$$\lambda(1236+1245)+\lambda'(3146+3245)+\lambda''(5146+5236)=0. \tag{14}$$

We have thus three linear equations, which written in a slightly different form are

$$\lambda\cdot 1235+\lambda'\cdot 3451+\lambda''\cdot 5613=0, \tag{15}$$
$$\lambda(1236+1245)+\lambda'(3452+3461)+\lambda''(5614+5623)=0, \tag{16}$$
$$\lambda\cdot 1246+\lambda'\cdot 3462+\lambda''\cdot 5624=0, \tag{17}$$

and thence eliminating λ, λ', λ'', we have

$$\begin{vmatrix} 1235, & 3451, & 5613 \\ 1236+1245, & 3452+3461, & 5614+5623 \\ 1246, & 3462, & 5624 \end{vmatrix}=0, \tag{18}$$

which is the required result.

It may be remarked that the second and third column are obtained from the first by operating on it with Δ, $\frac{1}{2}\Delta^2$, if $\Delta=2\delta_1+4\delta_2+6\delta_3$.

Or say the result is

$$(1,\Delta,\tfrac{1}{2}\Delta^2)\begin{vmatrix} 1235 \\ 3451 \\ 5613 \end{vmatrix}=0. \tag{19}$$

In like manner for the system

$$\begin{array}{cccccc} x^5, & x^4y, & x^3y^2, & x^2y^3, & xy^4, & y^5 \\ a, & b, & c, & d, & e, & f \\ a', & b', & c', & d', & e', & f' \\ a'', & b'', & c'', & d'', & e'', & f'' \\ a''', & b''', & c''', & d''', & e''', & f''' \end{array} \tag{18}$$

if the columns are

$$\begin{array}{cc|cc|cc|cc} a & b, & a' & b', & a'' & b'', & a''' & b''' \\ b & c, & b' & c', & b'' & c'', & b''' & c''' \\ c & d, & c' & d', & c'' & d'', & c''' & d''' \\ d & e, & d' & e', & d'' & e'', & d''' & e''' \\ e & f, & e' & f', & e'' & f'', & e''' & f''' \end{array} \tag{19}$$

$= 1, 2,\ 3, 4,\ 5, 6,\ 7, 8$; then the result is

$$(1, \Delta, \tfrac{1}{2}\Delta^2, \tfrac{1}{6}\Delta^3) \begin{vmatrix} 12357 \\ 34571 \\ 56713 \\ 78135 \end{vmatrix} = 0, \tag{20}$$

where

$$\Delta = 2\delta_1 + 4\delta_3 + 6\delta_5 + 8\delta_7. \tag{21}$$

392. On the Conics which pass through Two Given Points and touch Two Given Lines.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. VIII (1867), pp. 211–219.]

392.

Let $x = 0$, $y = 0$ be the equations of the given lines; $z = 0$ the equation of the line joining the given points.

We may, to fix the ideas, imagine the implicit constants so determined that $x + y + z = 0$ shall be the equation of the line infinity.

Take $x - my = 0$, $x - ny = 0$ as the equations of the lines which by their intersection with $z = 0$ determine the given points.

The equation of the conic is

$$(x - my)(x - ny) + 2\{x + y\sqrt{(mn)}\}\sqrt{(yz)} = x + y\sqrt{(mn)} + \gamma z, \tag{1}$$

or, what is the same thing,

$$(x - my)(x - ny) + 2\{x + y\sqrt{(mn)}\}\gamma z + \gamma^2 z^2 = 0, \tag{2}$$

so that there are two distinct series of conics according as $\sqrt{(mn)}$ is taken with the positive or the negative sign.

The equation of the chord of contact is

$$x + y\sqrt{(mn)} + \gamma z = 0, \tag{3}$$

which meets $z = 0$ in the point $[x + y\sqrt{(mn)} = 0,\ z = 0]$, that is in one of the centres of the involution formed by the lines $(x = 0,\ y = 0)$, $(x - my = 0,\ x - ny = 0)$.

It is to be observed that the conic is only real when mn is positive, that is (the lines and points being each real) the two points must be situate in the same region or in opposite regions of the four regions formed by the two lines: there are however other real cases; e.g. if the lines $x = 0$, $y = 0$ are real, but the quantities m, n are conjugate imaginaries; included in this we have the circles which touch two real lines.

To fix the ideas I take m and n each positive and $mn > 1$; also I attend first to the series where $\sqrt{(mn)}$ is taken positively.

At the points where the conic meets infinity, we have

$$\{\sqrt{(m)} + \sqrt{(n)}\}\sqrt{(xy)} = x + y\sqrt{(mn)} - \gamma(x + y), \tag{4}$$

which gives two coincident points, that is the conic is a parabola, if

$$(1 - \gamma)\{\sqrt{(mn)} - \gamma\} = \tfrac{1}{4}\{\sqrt{(m)} + \sqrt{(n)}\}^2, \tag{5}$$

that is

$$\gamma = \tfrac{1}{2}\{1 + \sqrt{(mn)} \pm \sqrt{(1 + m)(1 + n)}\}, \tag{6}$$

where it is to be noticed that

$$\gamma = \tfrac{1}{2}[1 + \sqrt{(mn)} + \sqrt{\{(1 + m)(1 + n)\}}] \tag{7}$$

is a positive quantity greater than $\sqrt{(mn)}$, say $\gamma = p$;

$$\gamma = \tfrac{1}{2}[1 + \sqrt{(mn)} - \sqrt{\{(1 + m)(1 + n)\}}] \tag{8}$$

is a negative quantity, say $\gamma = -q$, q being positive.

The order of the lines is as shown in fig. 1, see plate facing p. 52.

$\gamma = -\infty$ to $\gamma = -q$, curve is ellipse;

$\gamma = -q$, parabola P_1;

$\gamma = -q$ to $\gamma = p$, curve is hyperbola;

$\gamma = p$, parabola P_2;

$\gamma = p$ to $\gamma = \infty$, ellipse.

Resuming the equation

$$(x-my)(x-ny)+2\{x+y\sqrt{(mn)}\}yz+\gamma^2z^2=0, \tag{9}$$

the coefficients are

$$(a,b,c,f,g,h)=(1,\ mn,\ \gamma^2,\ \gamma\sqrt{(mn)},\ \gamma,\ -\tfrac{1}{2}(m+n)), \tag{10}$$

and thence the inverse coefficients are

$$\begin{aligned}(A,B,C,F,G,H)=(0,\ 0,\ -\tfrac{1}{4}(m-n)^2,\ -\tfrac{1}{4}\gamma\{\sqrt{(m)}+\sqrt{(n)}\}^2,&\\ -\tfrac{1}{4}\gamma\sqrt{(mn)}\{\sqrt{(m)}+\sqrt{(n)}\}^2,\ \tfrac{1}{4}\gamma^2\{\sqrt{(m)}+\sqrt{(n)}\}^2).&\end{aligned} \tag{11}$$

$K=-\frac{1}{4}\gamma^2\{\sqrt{(m)}+\sqrt{(n)}\}^2$,
or, omitting a factor, the inverse coefficients are

$$(A,B,C,F,G,H)=(0,\ 0,\ -\gamma\{\sqrt{(m)}-\sqrt{(n)}\}^2,\ 1,\ \sqrt{(mn)},\ -\gamma). \tag{12}$$

Considering the line $\lambda x + \mu y + \nu z = 0$, the coordinates of the pole of this line are

$$X : Y : Z = -\gamma\mu + \sqrt{(mn)}\nu : -\gamma\lambda + \nu : \sqrt{(mn)}\lambda + \mu + \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2\nu, \tag{12}$$

or (what is the same thing) introducing the arbitrary coefficient k, we have

$$kx + y\mu - \sqrt{(mn)} = 0, \tag{13}$$

$$ky + \gamma\lambda - \mu = 0, \tag{14}$$

$$kz - \lambda\sqrt{(mn)} - \mu - \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2\nu = 0; \tag{15}$$

the first two equations give

$$k : \gamma : -1 = \nu\{\mu - \lambda\sqrt{(mn)}\} : \nu\{\gamma\sqrt{(mn)} - \lambda\} : \lambda x - \mu y, \tag{16}$$

that is

$$k = \frac{-\nu\{\mu - \lambda\sqrt{(mn)}\}}{\lambda x - \mu y}, \quad \gamma = \frac{-\nu\{\gamma\sqrt{(mn)} - x\}}{\lambda x - \mu y}, \tag{17}$$

or, substituting this value of γ in the third equation,

$$\frac{\nu\{\mu - \lambda\sqrt{(mn)}\}(x - y\sqrt{(mn)})}{\lambda x - \mu y} + \frac{\{\mu + \lambda\sqrt{(mn)}\}(\lambda x - \mu y)}{\lambda x - \mu y} + \frac{\tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2\nu(x - y\sqrt{(mn)})}{\lambda x - \mu y} = 0, \tag{18}$$

that is

$$(\lambda x - \mu y)^2 \cdot \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 + \{x - y\sqrt{(mn)}\}(\lambda x - \mu y)\{\mu + \lambda\sqrt{(mn)}\} + z\{x - y\sqrt{(mn)}\}\nu\{\mu - \lambda\sqrt{(mn)}\} = 0, \tag{19}$$

which is the equation of the curve, the locus of the pole of the line $\lambda x + \mu y + \nu z = 0$ in regard to the conic $(x - my)(x - ny) + 2\{x + y\sqrt{(mn)}\}yz + \gamma^2 z^2 = 0$.

In particular, if $\lambda = \mu = \nu = 1$, then for the coordinates of the centre of the conic, we have

$$X : Y : Z = -\gamma + \sqrt{(mn)} : -\gamma + 1 : \sqrt{(mn)} + 1 + \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2; \tag{18}$$

and for the locus of the centre,

$$(X - Y)^2 \cdot \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 + (x - y)\{x - y\sqrt{(mn)}\}\{1 + \sqrt{(mn)}\} \\ + z\{x - y\sqrt{(mn)}\}\{1 - \sqrt{(mn)}\} = 0, \tag{19}$$

so that the locus is a conic, and it is obvious that this conic is a hyperbola.

Putting for greater simplicity

$$x - y = X, \quad x - y\sqrt{(mn)} = Y, \quad z = Z, \tag{20}$$

the equation of the curve of centres is

$$Z^2 \cdot \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 + XY\{1 + \sqrt{(mn)}\} + YZ\{1 - \sqrt{(mn)}\} = 0, \tag{21}$$

or, writing this under the form

$$Y[Z\{1 + \sqrt{(mn)}\} + Z\{1 - \sqrt{(mn)}\}] + \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 Z^2 = 0, \tag{22}$$

the equation is

$$XY + ZQ = 0, \tag{22}$$

where

$$X = x - y, \tag{23}$$

$$Y = x - y\sqrt{(mn)}, \tag{24}$$

$$Q = \frac{2}{\{\sqrt{(m)} - \sqrt{(n)}\}^2}[Z\{1 + \sqrt{(mn)}\}(x - y) + \{1 - \sqrt{(mn)}\}z]; \tag{25}$$

these values give

$$x - y = X, \tag{26}$$

$$x - y\sqrt{(mn)} = Y, \tag{27}$$

$$\{1 - \sqrt{(mn)}\}z = \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 Q + 2\{1 + \sqrt{(mn)}\}X, \tag{28}$$

or, what is the same thing,

$$\{1 - \sqrt{(mn)}\}x = -\sqrt{(mn)}X + Y, \tag{29}$$

$$\{1 - \sqrt{(mn)}\}y = -X + Y, \tag{30}$$

$$\{1 - \sqrt{(mn)}\}z = 2\{1 + \sqrt{(mn)}\}X + \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 Q. \tag{31}$$

whence also

$$\{1 - \sqrt{(mn)}\}(x + y + z) = \{1 + \sqrt{(mn)}\}Z + 2Y + \{\sqrt{(m)} - \sqrt{(n)}\}^2 Q, \tag{32}$$

or the equation of the line infinity is

$$\{1 + \sqrt{(mn)}\}X + 2Y + \{\sqrt{(m)} - \sqrt{(n)}\}^2 Q = 0, \tag{33}$$

a formula which may be applied to finding the asymptotes and thence the centre of the conic $YQ+X^2 = 0$.

In fact we have identically

$$\{2kx + 2ky - (2k + 1)z\}^2 - (1 + 4k)(2kx - z)^2 = 4k^2(x + y + z)^2 - 4k(1 + 4k)(kx^2 + yz), \tag{34}$$

that is

$$-4k(1 + 4k)(kx^2 + yz) = \{2kx + 2ky - (2k + 1)z\}^2 - (1 + 4k)(2kx - z)^2 - 4k^2(x + y + z)^2, \tag{35}$$

which, if $x + y + z = 0$ is the equation of the line infinity, puts in evidence the asymptotes of the conic $kx^2 + yz = 0$.

Hence writing αx, βy, γz in the place of x, y, z respectively, and $k = k'$, that is, $k = \dfrac{\beta\gamma}{\alpha^2}k'$, we have

$$\begin{aligned}-4\beta\gamma k'(\alpha^2+4\beta\gamma k')(k'x^2+yz) = \{&2\alpha\gamma k'x + 2\beta^2 k'y - (2\beta\gamma k'+\alpha^2)z\}^2 \\ &- (\alpha^2+4\beta\gamma k')(2\beta\gamma k'x-\alpha z)^2 - 4\beta^2\gamma^2 k'^2(\alpha x+\beta y+\gamma z)^2. \quad (31)\end{aligned}$$

Now writing X, Y, Q in the place of x, y, z; $k' = 1$, and $\alpha = \{1+\sqrt{(mn)}\}$, $\beta = 2$, $\gamma = \{\sqrt{(m)}-\sqrt{(n)}\}^2$, we have

$$\begin{aligned}&-16[\{1+\sqrt{(mn)}\}^2+8\{\sqrt{(m)}-\sqrt{(n)}\}^2](YQ+X^2) \\ &\quad = [4\{1+\sqrt{(mn)}\}X + 8Y - (4\{\sqrt{(m)}-\sqrt{(n)}\}^2+\{1+\sqrt{(mn)}\}^2)Q]^2 \\ &\quad\quad - [\{1+\sqrt{(mn)}\}^2+8\{\sqrt{(m)}-\sqrt{(n)}\}^2][4X-\{1+\sqrt{(mn)}\}Q]^2 \\ &\quad\quad - 16[\{1+\sqrt{(mn)}\}X+2Y+\{\sqrt{(m)}-\sqrt{(n)}\}^2Q]^2, \quad (32)\end{aligned}$$

and the asymptotes are

$$\begin{aligned}&4\{1+\sqrt{(mn)}\}X+8Y-[4\{\sqrt{(m)}-\sqrt{(n)}\}^2+\{1+\sqrt{(mn)}\}^2]Q \\ &\quad = \pm\sqrt{\{1+\sqrt{(mn)}\}^2+8\{\sqrt{(m)}-\sqrt{(n)}\}^2}\,[4X-\{1+\sqrt{(mn)}\}Q]. \quad (33)\end{aligned}$$

At the centre

$$4(1+\sqrt{(mn)})X+8Y-[4\{\sqrt{(m)}-\sqrt{(n)}\}^2+\{1+\sqrt{(mn)}\}^2]Q=0, \quad (34)$$

$$4X-\{1+\sqrt{(mn)}\}Q=0, \quad (35)$$

but the first equation is

$$\{1+\sqrt{(mn)}\}[4X-Q\{1+\sqrt{(mn)}\}]+8Y-4\{\sqrt{(m)}-\sqrt{(n)}\}^2Q=0, \quad (36)$$

so that we have

$$4X=\{1+\sqrt{(mn)}\}Q, \quad (37)$$

$$2Y=\{\sqrt{(m)}-\sqrt{(n)}\}^2Q, \quad (38)$$

the first of these is

$$2\{\sqrt{(m)}-\sqrt{(n)}\}^2(x-y)-\{1+\sqrt{(mn)}\}^2(x-y)-(1-mn)z=0, \quad (39)$$

and the two together give

$$2Z\{\sqrt{(m)}-\sqrt{(n)}\}^2-\{1+\sqrt{(mn)}\}Y=0, \quad (40)$$

so that we have

$$2\{\sqrt{(m)}-\sqrt{(n)}\}^2(x-y)-\{1+\sqrt{(mn)}\}(x-y\sqrt{(mn)})-(1-mn)z=0. \quad (41)$$

393. On the Conics which touch Three Given Lines, &c.

[From the *Proceedings of the London Mathematical Society*, vol. IV (1871), pp. 114–115.]

393.

The problem of determining the conics which touch three given lines and satisfy some other condition is considered. When the three lines are given, the conic is determined by three tangential conditions. If additional conditions are given, such as passing through a point or touching a fourth line, the conic is completely determined.

The analytical treatment of this problem leads to interesting equations. Let the three given lines be

$$\lambda x + \mu y + \nu z = 0, \quad \lambda' x + \mu' y + \nu' z = 0, \quad \lambda'' x + \mu'' y + \nu'' z = 0. \tag{1}$$

The condition that a conic touches these three lines can be expressed by means of the tangential equation of the conic.

[Plate — see scan: I facing page 52: Figure illustrating the conics touching three given lines.]

394. On a Locus in relation to the Triangle.

[From the *Proceedings of the London Mathematical Society*, vol. IV (1871), pp. 115–121.]

394.

The problem of finding the locus of a point related in a given manner to a triangle leads to many interesting curves.

Let the triangle be ABC, and let the point P be such that its distances from the sides have a given relation. The locus of P is in general a curve of higher order.

Consider the case where the point is the intersection of lines drawn from the vertices with given directions. The locus can be represented by the equation

$$\begin{vmatrix} A & H & G \\ H & B & F \\ G & F & C \end{vmatrix} = 0. \tag{1}$$

Consider the case where the point is such that the perpendiculars from it on the sides of the triangle have their feet in a line.

Let the sides of the triangle be $x = 0$, $y = 0$, $z = 0$, and let the Absolute (the conic in regard to which the perpendiculars are drawn) be

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0. \tag{2}$$

The condition that the feet of the perpendiculars lie in a line is that the point (x, y, z) satisfies a certain cubic equation.

that is, the locus is a cubic curve passing through the vertices of the triangle.

The condition

$$\frac{abc}{af^2} \cdot \frac{1}{bg^2} \cdot \frac{1}{ch^2} = \frac{1}{fgh} \tag{3}$$

is the condition in order that the function

$$\frac{\xi^2}{a} + \frac{\eta^2}{b} + \frac{\zeta^2}{c} \tag{4}$$

may break up into linear factors; the function in question is

$$\frac{bc}{f^2} + \frac{ca}{g^2} + \frac{ab}{h^2}, \tag{5}$$

which is

$$\left(\frac{bc}{f}, \frac{ca}{g}, \frac{ab}{h} \mid x, y, z\right)^2 = (a, b, c, f, g, h \mid x, y, z)^2 + 2\left(\frac{f}{a}yz + \frac{g}{b}zx + \frac{h}{c}xy\right), \tag{6}$$

so that the condition is, that the conic

$$(a, b, c, f, g, h \mid x, y, z)^2 + 2\left(\frac{f}{a}yz + \frac{g}{b}zx + \frac{h}{c}xy\right) = 0, \tag{7}$$

(which is a certain conic passing through the intersections of the Absolute $(a, b, c, f, g, h \mid x, y, z)^2 = 0$, and of the locus conic $\frac{f}{a}yz + \frac{g}{b}zx + \frac{h}{c}xy = 0$) shall be a pair of lines.

Writing the equation of the conic in question under the form

$$\left(\frac{bc}{f}, \frac{ca}{g}, \frac{ab}{h} \mid x, y, z\right)^2, \tag{8}$$

the inverse coefficients A', B', C', F', G', H' of this conic, are

$$\frac{A'bc}{f^2} = \frac{B'ca}{g^2} = \frac{C'ab}{h^2} = \frac{abc}{fgh}, \tag{9}$$

so that we have $F' : G' : H' = F : G : H$.

Hence, if in regard to this new conic we form the reciprocal of the triangle ($x = 0,\ y = 0,\ z = 0$), and join the corresponding angles of the two triangles, the joining lines meet in a point which is the same point as is obtained by the like process from the triangle and its reciprocal in regard to the Absolute.

But I do not further pursue this part of the theory.

It is to be noticed that the conic

$$\frac{A}{f}yz + \frac{B}{g}zx + \frac{C}{h}xy = 0, \tag{10}$$

contains the angles of the reciprocal triangle, and is thus in fact the conic in which are situate the angles of the two triangles.

For the coordinates of one of the angles of the reciprocal triangle are (A, H, G); we should therefore have

$$\frac{A}{f}HG + \frac{B}{g}GA + \frac{C}{h}AB = 0, \tag{11}$$

which is $\frac{1}{f}(GH \cdot gh + BG \cdot hf + GH \cdot fg) = 0$, or attending only to the second factor and writing $GH = Kf + AF$, the condition is

$$Kfgh + AFgh + BGhf + CHfg = 0, \tag{12}$$

or substituting for K, A, B, C, F, G, H their values and reducing, this is satisfied: hence the three angles of the reciprocal triangle lie on the conic in question.

Partially recapitulating the foregoing results, we see in the case where the Absolute is not a point-pair, that the locus of a point such that the perpendiculars from it on the sides of the triangle have their feet in a line, is in general a cubic curve passing through the angles of the triangle: if, however, the condition

$$\frac{abc}{af^2} \cdot \frac{1}{bg^2} \cdot \frac{1}{ch^2} = \frac{1}{fgh} \tag{13}$$

be satisfied, that is, if the triangle be such that the angles thereof and of the reciprocal triangle lie in a conic (or, what is the same thing, if the sides touch a conic) then the cubic locus breaks up into the line

$$\frac{f}{a}x + \frac{g}{b}y + \frac{h}{c}z = 0, \tag{14}$$

which is the line through the points of intersection of the corresponding sides of the two triangles, and into the conic

$$\frac{A}{f}yz + \frac{B}{g}zx + \frac{C}{h}xy = 0, \tag{15}$$

which is the conic through the angles of the two triangles.

The question arises, given a conic (the Absolute) to construct a triangle such that its angles, and the angles of the reciprocal triangle in regard to the given conic, lie in a conic.

I suppose that two of the angles of the triangle are given, and I enquire into the locus of the remaining angle.

To fix the ideas, let A, B, C be the angles of the triangle, A', B', C' those of the reciprocal triangle; and let the angles A and B be given. We have to find the locus of the point C: I observe however, that the lines AA', BB', CC' meet in a point O, and I conduct the investigation in such manner as to obtain simultaneously the loci of the two points C and O.

The lines $C'B'$, $C'A'$ are the polars of A, B respectively, let their equations be $x=0$, and $y=0$, and let the equation of the line AB be $z=0$; this being so, the equation of the given conic will be of the form

$$(a, b, c, 0, 0, h \mid x, y, z)^2 = 0. \tag{16}$$

I take (α, β, γ) for the coordinates of C and (x, y, z) for those of O; the coordinates of either of these points being of course deducible from those of the other.

Observing that the inverse coefficients are $(bc,\ ca,\ ab-h^2,\ 0,\ 0,\ -ch)$, we find

Coordinates of A are $(b, -h, 0)$,

B „ $(-h, a, 0)$.

The points A' and B' are then given as the intersections of AO with $C'A'$ $(y=0)$ and of BO with $C'B'$ $(x=0)$; we find

Moreover,

Coordinates of A' are $(h\alpha+b\beta,\ a\alpha+h\beta,\ h\gamma)$,

B' „ $(h\alpha+b\beta,\ a\alpha+h\beta,\ h\gamma)$,

Coordinates of C are $(0, 0, 1)$,

C' „ (α, β, γ).

The six points A, B, C, A', B', C' are to lie in a conic; the equations of the lines CA, CB, AB are $hX+bY=0$, $aX+hY=0$, $Z=0$, and hence the equation of a conic passing through the points C, A, B is

$$\frac{hX+bY}{Z}\cdot\frac{aX+hY}{Z} = 0. \tag{17}$$

Hence, making the conic pass through the remaining points A', B', C, we find

$$\frac{a\alpha(h\alpha+b\beta)}{L}+\frac{h(h\alpha+b\beta)}{M}+\frac{h\gamma}{N}=0,\quad \frac{h(a\alpha+h\beta)}{L}+\frac{b(a\alpha+h\beta)}{M}+\frac{h\gamma}{N}=0, \tag{18}$$

and eliminating the L, M, N, we find

$$\begin{vmatrix} 1 & a & 1 \\ 1 & h & 1 \\ 1 & h & 1 \\ h\alpha+b\beta & a\alpha+h\beta & h\gamma \\ a\alpha+h\beta & h\alpha+b\beta & z \end{vmatrix} = 0, \tag{19}$$

or developing and reducing, this is

$$\frac{ab-h^2}{\gamma}Y^2=\frac{1}{a\alpha+h\beta}+\frac{1}{h\alpha+b\beta}-\frac{h\alpha\beta}{a\alpha+h\beta}-\frac{ab}{h\alpha+b\beta}, \tag{20}$$

that is

$$\frac{ab-h^2}{\gamma}Y^2=(hb,-ab,ha \mid a\alpha+h\beta,h\alpha+b\beta)^2=0. \tag{21}$$

We have still to find the relation between (α,β,γ) and (x,y,z); this is obtained by the consideration that the line $A'B'$, through the two points A', B' the coordinates of which are known in terms of (α,β,γ), is the polar of the point C, the coordinates of which are (x,y,z).

The equation of $A'B'$ is thus obtained in the two forms

$$(a\alpha+h\beta)Z+(h\alpha+b\beta)Y-\frac{(a\alpha+h\beta)(h\alpha+b\beta)}{ch\gamma}X=0, \tag{22}$$

and

$$(ax+hy)X+(hx+by)Y+cz=0, \tag{23}$$

and comparing these, we have

$$X:Y:Z=\alpha:\beta:\gamma=x:y:\frac{-(a\alpha+h\beta)(h\alpha+b\beta)}{ch\gamma}, \tag{24}$$

or what is the same thing

$$\alpha:\beta:\gamma=X:Y:\frac{-(a\alpha+h\beta)(h\alpha+b\beta)}{ch\gamma}=x:y:\frac{-(ax+hy)(hx+by)}{chz} \tag{25}$$

(where it is to be observed that the equation $\alpha:\beta=x:y$ is the verification of the theorem that the lines AA', BB', CC' meet in a point O).

We may now from the above found relation eliminate either the (α, β, γ) or the (x, y, z); first eliminating the (α, β, γ), we find

$$\frac{ab-h^2}{\gamma}Y^2 = \frac{h}{a}\frac{1}{ax+hy} + \frac{1}{b}\frac{1}{hx+by} - \frac{hab}{(ax+hy)(hx+by)}, \tag{24}$$

where

$$\frac{Y}{Z} = \frac{(a\alpha+h\beta)(h\alpha+b\beta)}{ch\gamma^2} = \frac{(ax+hy)(hx+by)}{chz^2}, \tag{25}$$

or, completing the elimination,

$$\frac{ab-h^2}{\gamma}\frac{(ax+hy)^2(hx+hy)^2}{chz^2} = (hb, -ab, ha \mid ax+hy, hx+by)^2 = 0, \tag{26}$$

which is a quartic curve having a node at each of the points

$$(z=0,\ ax+hy=0), \quad (z=0,\ hx+by=0), \quad (ax+hy=0,\ hx+by=0), \tag{27}$$

that is, at each of the points B, A, C.

The right-hand side of the foregoing equation is

$$= -(ab-h^2)(ha, ah, hh \mid x, y)^2 = -(ab-h^2)h(ax^2+by^2+\tfrac{2h}{ab}xy), \tag{28}$$

so that the equation may also be written

$$(ax+hy)^2(hx+by)^2 + ch^2z^2\left(ax^2+by^2+\tfrac{2h}{ab}xy\right) = 0. \tag{29}$$

Secondly, to eliminate the (x, y, z), we have

$$\frac{ab-h^2}{\gamma}Y^2 = \frac{1}{h}\frac{1}{a\alpha+h\beta} + \frac{1}{a}\frac{1}{h\alpha+b\beta}, \tag{30}$$

where

$$\frac{Y}{Z} = \frac{(a\alpha+h\beta)(h\alpha+b\beta)}{ch\gamma^2}, \tag{31}$$

or, completing the elimination,

$$(ab-h^2)ch\gamma^2 = (hb, -ab, ha \mid a\alpha+h\beta, h\alpha+b\beta)^2, \tag{32}$$

that is

$$= -(ab-h^2)h(a\alpha^2+b\beta^2+\tfrac{2h}{ab}\alpha\beta), \tag{33}$$

$$(a, b, c, 0, 0, h \mid \alpha, \beta, \gamma)^2 = 0. \tag{34}$$

Writing (x, y, z) in place of (α, β, γ), the locus of the point is the conic

$$(a, b, c, 0, 0, h \mid x, y, z)^2 = 0, \tag{35}$$

which is a conic intersecting the Absolute $(a, b, c, 0, 0, h \mid x, y, z)^2 = 0$, at its intersections with the lines $x = 0$, $y = 0$, that is the lines CB and CA'.

In regard to this new conic, the coordinates of the pole of $C'B'$ $(x = 0)$ are at once found to be $(-h, a, 0)$, that is, the pole of $C'B'$ is B; and similarly the coordinates of the pole of $C'A'$ $(y = 0)$ are $(b, -h, 0)$, that is, the pole of $C'A'$ is A.

We may consequently construct the conic the locus of O, viz. given the Absolute and the points A and B, we have $C'A'$ the polar of B, meeting the Absolute in two points (a_1, a_2), and $C'B'$ the polar of A meeting the Absolute in the points (b_1, b_2); the lines $C'A'$ and $C'B'$ meet in C'.

This being so, the required conic passes through the points a_1, a_2, b_1, b_2, the tangents at these points being Aa_1, Aa_2, Bb_1, Bb_2 respectively; eight conditions, five of which would be sufficient to determine the conic.

It is to be remarked that the lines $C'B'$, $C'A'$ (which in regard to the Absolute are the polars of A, B respectively) are in regard to the required conic the polars of B, A respectively.

The conic the locus of O being known, the point may be taken at any point of this conic, and then we have A' as the intersection of $C'A'$ with AO, B' as the intersection of $C'B'$ with BO.

to the Absolute, the point so obtained being a point on the line CO.

To each position of O on the conic locus, there corresponds of course a position of C; the locus of C is, as has been shown, a quartic curve having a node at each of the points C, A, B.

The foregoing conclusions apply of course to spherical figures; we see therefore that on the sphere the locus of a point such that the perpendiculars let fall on the sides of a given spherical triangle have their feet in a line (great circle), is a spherical cubic.

If, however, the spherical triangle is such that the angles thereof and the poles of the sides (or, what is the same thing, the angles of the polar triangle) lie on a spherical conic; then the cubic locus breaks up into a line (great circle), which is in fact the circle having for its pole the point of intersection of the perpendiculars from the angles of the triangle on the opposite sides respectively, and into the before-mentioned spherical conic.

395. Investigations in connexion with Casey's Equation.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XII (1873), pp. .]

395.

The equation of Casey relates to the condition for four circles to touch a fifth circle. If ρ_1, ρ_2, ρ_3, ρ_4 are the radii of four circles touching a fifth circle of radius ρ, and t_{12}, t_{13}, etc. are the distances between their centres, then Casey's equation is

$$\begin{vmatrix} 0 & t_{12}^2 & t_{13}^2 & t_{14}^2 & 1 \\ t_{21}^2 & 0 & t_{23}^2 & t_{24}^2 & 1 \\ t_{31}^2 & t_{32}^2 & 0 & t_{34}^2 & 1 \\ t_{41}^2 & t_{42}^2 & t_{43}^2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{vmatrix} = 0. \tag{1}$$

If however the circles all touch the fifth circle externally, the equation takes a simpler form. Writing $t_{ij}^2 = (\rho_i + \rho_j)^2$ for external contact, or $t_{ij}^2 = (\rho_i - \rho_j)^2$ for internal contact, the determinant condition simplifies.

The proof depended on the properties of the circles of inversion. By inverting the system with respect to a circle whose centre is at one of the points of contact, the circles become parallel lines, and the condition is evident.

equation of
The equation may be written in the form

$$\begin{vmatrix} \rho_1 & \rho_2 & \rho_3 & \rho_4 \\ 1 & 1 & 1 & 1 \\ t_{23}^2 & t_{13}^2 & t_{12}^2 & \cdots \\ \cdots & \cdots & \cdots & \cdots \end{vmatrix} = 0. \tag{2}$$

and substituting the values of t_{ij}^2 in terms of the radii, we obtain a relation which must be satisfied.

The condition for four circles to touch a fifth circle may also be expressed in terms of the curvatures. If the circles have curvatures κ_1, κ_2, κ_3, κ_4 (where $\kappa = 1/\rho$), and the distances between centres are t_{ij}, then the condition is that a certain determinant involving these quantities shall vanish.

The general theory of the contacts of circles may be applied to the determination of the circles which touch three given circles. The problem of Apollonius, to find a circle touching three given circles, has eight solutions, corresponding to the different combinations of external and internal contact.

The investigation of Casey's equation leads to many interesting results in the geometry of circles. The condition may be generalized to the case of spheres in three dimensions, and to higher dimensions.

The relation between the radii and the distances may be expressed in a symmetric form. Writing $u_{ij} = t_{ij}^2/\rho_i\rho_j$, the condition becomes a relation between the quantities u_{ij} and the ratios of the radii.

The equation of Casey is a fundamental result in the theory of the contacts of circles. It has been applied to the study of the configurations of circles in a plane, and to the investigation of the properties of systems of circles.

The proof of Casey's equation may be obtained by considering the inversion of the system of circles. If we invert with respect to a circle whose centre is at the point of contact of two of the circles, these circles become parallel lines, and the condition for the contact of the remaining circles with the inverted circle is easily established.

396. On a certain Envelope depending on a Triangle inscribed in a Circle.

[From the *Proceedings of the London Mathematical Society*, vol. IV (1871), pp. 122–127.]

396.

Let a triangle be inscribed in a circle, and consider the envelope of a line connected with the triangle by a given relation.

We may take the circle to be $x^2 + y^2 = 1$, and the vertices of the triangle to be points on this circle.

and I proceed to find the equation of the envelope.

Let the vertices of the triangle be represented parametrically by the angles α, β, γ, so that the coordinates are $(\cos\alpha, \sin\alpha)$, $(\cos\beta, \sin\beta)$, $(\cos\gamma, \sin\gamma)$.

The equation of the line joining the points with parameters β and γ is

$$x\cos\tfrac{1}{2}(\beta+\gamma) + y\sin\tfrac{1}{2}(\beta+\gamma) = \cos\tfrac{1}{2}(\beta-\gamma). \tag{1}$$

We find the equation by eliminating the parameters between the equations of the line and the condition of envelopment.

Let the line be $\lambda x + \mu y + \nu = 0$, and let the condition of envelopment be that the line is connected with the triangle by a given relation, e.g. that it passes through a fixed point, or that it touches a given conic.

where the coefficients of the line equation are functions of the parameters determining the triangle.

The envelope of the line is obtained by eliminating the parameters between the equation of the line and the condition that the line touches the envelope. This leads to a curve of higher class.

I find the envelope to be a curve of the sixth class, having cusps and nodes depending on the particular relation assumed.

The equation of the envelope may be obtained by eliminating the parameters between the equations

$$\lambda\cos\alpha+\mu\sin\alpha+\nu=0,\quad \lambda\cos\beta+\mu\sin\beta+\nu=0,\quad \lambda\cos\gamma+\mu\sin\gamma+\nu=0, \tag{2}$$

and the condition of envelopment.

I write the equation in the form

$$(a, b, c, d, e, f \mid x, y, z)^6 = 0, \tag{3}$$

where the coefficients are functions of the parameters determining the relation between the line and the triangle.

The class of the envelope is six, and the curve has in general nine cusps and four nodes. The particular case where the envelope degenerates into simpler curves is of special interest.

Original scan reference pages

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I suppose in particular that the two conics are

$$x^2 + my^2 - 1 = 0,$$
$$mx^2 + \quad y^2 - 1 = 0,$$

the equation of the quartic is

$$4(2k+1)^2 m^2 (x^2 + my^2 - 1)(mx^2 + y^2 - 1) - \{(m^2 + m)(x^2 + y^2) - m^2 - 1\}^2 = 0;$$

or putting $\lambda = \dfrac{(m+1)^2}{4(2k+1)^2}$, this is

$$\lambda\left(x^2 + y^2 - \frac{m^2+1}{m^2+m}\right)^2 - (x^2 + my^2 - 1)(mx^2 + y^2 - 1) = 0.$$

To fix the ideas, suppose that m is positive and >1, so that each of the conics is an ellipse, the major semi-axis being $=1$, and the minor semi-axis being $= \dfrac{1}{\sqrt{(m)}}$. For any real value of k the coefficient λ is positive, and it may accordingly be assumed that λ is positive.

We have $\dfrac{m^2+1}{m(m+1)} > \dfrac{1}{m} < 1$, or the radius of the circle is intermediate between the semi-axes of the ellipses, hence the points of contact on each ellipse are real points.

Writing for shortness

$$\alpha = \frac{m^2+1}{m^2+m},$$

the equation is

$$(x^2 + my^2 - 1)(mx^2 + y^2 - 1) - \lambda(x^2 + y^2 - \alpha)^2 = 0.$$

For the points on the axis of x, we have

$$(x^2 - 1)(mx^2 - 1) - \lambda(x^2 - \alpha)^2 = 0,$$

that is

$$(m - \lambda)x^4 + \{-(1+m) + 2\lambda\alpha\}x^2 + (1 - \lambda\alpha^2) = 0,$$

and thence

$$(m-\lambda)x^2 = \tfrac{1}{2}(1+m) - \lambda\alpha \pm \tfrac{1}{2}\sqrt{\{(m-1)^2 + 4\lambda(1-\alpha)(1-m\alpha)\}},$$

or, substituting for α its value, this is

$$(m-\lambda)x^2 = \tfrac{1}{2}(m+1) - \frac{\lambda\left(m + \dfrac{1}{m}\right)}{m+1} \pm \frac{\tfrac{1}{2}(m-1)}{m+1}\sqrt{\{(m+1)^2 - 4\lambda\}}.$$

Remarking that the values $\dfrac{(m+1)^2}{\left(m+\dfrac{1}{m}\right)^2}$, m, $\tfrac{1}{4}(m+1)^2$ are in the order of increasing magnitude, and considering successive values of λ; first the value $\lambda = \frac{1}{\alpha^2}, = \frac{(m+1)^2}{\left(m+\frac{1}{m}\right)^2}$, we have

$$(m-\lambda)\,x^2 = \tfrac{1}{2}(m+1) - \frac{m+1}{m+\frac{1}{m}} \pm \frac{\tfrac{1}{2}(m-1)\left(m-\frac{1}{m}\right)}{\left(m+\frac{1}{m}\right)}$$

$$= \frac{(m+1)\tfrac{1}{2}\left(m+\frac{1}{m}-2\right) \pm \tfrac{1}{2}(m-1)\left(m-\frac{1}{m}\right)}{\left(m+\frac{1}{m}\right)};$$

or observing that

$$(m+1)\left(m+\frac{1}{m}-2\right) = (m+1)\frac{1}{m}(m-1)^2 = \frac{1}{m}(m-1)(m^2-1) = (m-1)\left(m-\frac{1}{m}\right),$$

this is

$$(m-\lambda)\,x^2 = 0, \text{ or } \frac{(m-1)\left(m-\frac{1}{m}\right)}{m+\frac{1}{m}},$$

or, what is the same thing,

$$\frac{(m-1)(m^3+2m^2-1)}{m\left(m+\frac{1}{m}\right)^2}\,x^2 = 0, \text{ or } \frac{(m-1)\left(m-\frac{1}{m}\right)}{m+\frac{1}{m}}, \quad x^2 = 0, \text{ or } \frac{\left(m^2-\frac{1}{m^2}\right)m}{m^3+2m^2-1}.$$

The next critical value is $\lambda = m$. The curve here is

$$(x^2+my^2-1)(mx^2+y^2-1) - m(x^2+y^2-\alpha)^2 = 0,$$

that is

$$\begin{aligned} &m(x^4+y^4) + (1+m^2)\,x^2y^2 - (m+1)(x^2+y^2) + 1 \\ &- m(x^4+y^4) - \quad 2m \quad x^2y^2 + \quad 2m\alpha \quad (x^2+y^2) - m\alpha^2 = 0, \end{aligned}$$

that is

$$(m-1)^2\,x^2y^2 + (2m\alpha - m - 1)(x^2+y^2) + 1 - m\alpha^2 = 0,$$

or, substituting for α its value,

$$2m\alpha - m - 1 = \frac{2m^2+2}{m+1} - (m+1) = \frac{(m-1)^2}{m+1},$$

$$1 - m\alpha^2 = 1 - \frac{(m^2+1)^2}{m(m+1)^2} = -\frac{(m-1)^2(m^2+m+1)}{m(m+1)^2};$$

the equation is

$$x^2y^2 + \frac{1}{m+1}(x^2+y^2) - \frac{m^2+m+1}{m(m+1)^2} = 0,$$

or, as this may also be written,

$$\left(x^2+\frac{1}{m+1}\right)\left(y^2+\frac{1}{m+1}\right) - \frac{1}{m} = 0,$$

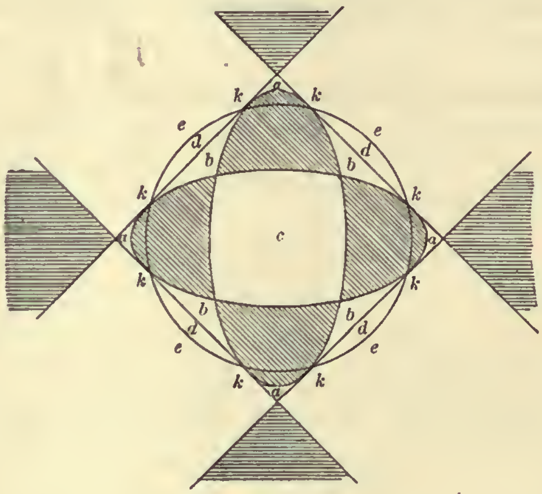
which has a pair of imaginary asymptotes parallel to the axis of x, and a like pair parallel to the axis of y, or what is the same thing, the curve has two isolated points at infinity, one on each axis.

The next critical value is $\lambda = \frac{1}{4}(m+1)^2$; the curve here reduces itself to the four lines

$$\left\{(x+y)^2 - \frac{m+1}{m}\right\}\left\{(x-y)^2 - \frac{m+1}{m}\right\} = 0;$$

and it is to be observed that when λ exceeds this value, or say $\lambda > \frac{1}{4}(m+1)^2$, the curve has no real point on either axis; but when $\lambda = \infty$, the curve reduces itself to $(x^2+y^2-\alpha)^2 = 0$, i.e. to the circle $x^2+y^2-\alpha = 0$ twice repeated, having in this special case real points on the two axes.

It is now easy to trace the curve for the different values of λ. The curve lies in every case within the unshaded regions of the figure (except in the limiting cases after-mentioned); and it also touches the two ellipses and the four lines at the eight points k, at which points it also cuts the circle; but it does not cut or touch the



four lines, the two ellipses, or the circle, except at the points k. Considering λ as varying by successive steps from 0 to ∞;

$\lambda = 0$, the curve is the two ellipses.

$\lambda < \dfrac{(m+1)^2}{\left(m+\dfrac{1}{m}\right)^2}$, the curve consists of two ovals, an exterior sinuous oval lying in the four regions a and the four regions b; and an interior oval lying in the region c.

$\lambda = \dfrac{(m+1)^2}{\left(m+\dfrac{1}{m}\right)^2}$, there is still a sinuous oval as above, but the interior oval has dwindled to a conjugate point at the centre.

$\lambda > \dfrac{(m+1)^2}{\left(m+\dfrac{1}{m}\right)^2} < m$; $\lambda = m$; $\lambda > m < \dfrac{(m+1)^2}{4}$; there is no interior oval, but only a sinuous oval as above; which, as λ increases, approaches continually nearer to the four sides of the square. For the critical value $\lambda = m$, there is no change in the general form, but the curve has for this value of λ, two conjugate points, one on each axis at infinity.

$\lambda = \frac{1}{4}(m+1)^2$, the curve becomes the four lines.

$\lambda > \frac{1}{4}(m+1)^2$, the curve lies wholly in the four regions a and the four regions e, consisting thereof of four detached sinuous ovals. As λ deviates less from the value $\frac{1}{4}(m+1)^2$, each oval approaches more nearly to the infinite trilateral formed by the side and infinite line-portions which bound the regions d, e to which the oval belongs. And as λ departs from the limit $\frac{1}{4}(m+1)^2$, and approaches to ∞, each sinuous oval approaches more nearly to the circular arc which separates the two regions d, e, which contains the sinuous oval.

Finally, $\lambda = 0$, the curve is the circle twice repeated.

390.

THEOREM RELATING TO THE FOUR CONICS WHICH TOUCH THE SAME TWO LINES AND PASS THROUGH THE SAME FOUR POINTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. VIII. (1867), pp. 162—167.]

THE sides of the triangle formed by the given points meet one of the given lines in three points, say P, Q, R; and on this same line we have four points of contact, say A_1, A_2, A_3, A_4; any two pairs, say A_1, A_2; A_3, A_4, form with a properly selected pair, say Q, R, out of the above-mentioned three points, an involution; and we have thus the three involutions

$$(A_1,\ A_2;\ \ A_3,\ A_4;\ \ Q,\ R),$$
$$(A_1,\ A_3;\ \ A_4,\ A_2;\ \ R,\ P),$$
$$(A_1,\ A_4;\ \ A_2,\ A_3;\ \ P,\ Q).$$

To prove this, let $x=0$, $y=0$ be the equations of the given lines, and take for the equations of the sides of the triangle formed by the given points

$$b\,x+a\,y-a\,b\ =0,$$
$$b'\,x+a'\,y-a'\,b'\ =0,$$
$$b''x+a''y-a''b''=0:$$

the equation of any one of the four conics may be written

$$\frac{Lab}{bx+ay-ab}+\frac{L'a'b'}{b'x+a'y-a'b'}+\frac{L''a''b''}{b''x+a''y-a''b''}=0,$$

and if this touches the axis of x, say at the point $x=\alpha$, then we must have

$$\frac{La}{x-a}+\frac{L'a'}{x-a'}+\frac{L''a''}{x-a''}=\frac{-K(x-\alpha)^2}{(x-a)(x-a')(x-a'')};$$

or, assuming as we may do, $K=-(a'-a'')(a''-a)(a-a')$, this gives

$$\begin{aligned} L\,a &= (a-\alpha)^2(a'-a''), \\ L'a' &= (a'-\alpha)^2(a''-a), \\ L''a'' &= (a''-\alpha)^2(a-a'). \end{aligned}$$

But in the same manner, if the conic touch the axis of y, say at the point $y=\beta$, we have

$$\begin{aligned} L\,b &= (b-\beta)^2(b'-b''), \\ L'b' &= (b'-\beta)^2(b''-b), \\ L''b'' &= (b''-\beta)^2(b-b'); \end{aligned}$$

and thence

$$\begin{aligned} & b(a-\alpha)^2(a'-a'') : b'(a'-\alpha)^2(a''-a) : b''(a''-\alpha)^2(a-a') \\ &= a(b-\beta)^2(b'-b'') : a'(b'-\beta)^2(b''-b) : a''(b''-\beta)^2(b-b'). \end{aligned}$$

Putting

$$\begin{aligned} P &= a\,b\,(a'-a'')(b'-b''), \\ P' &= a'b'(a''-a)(b''-b), \\ P'' &= a''b''(a-a')(b-b'), \end{aligned}$$

we have

$$(a-\alpha)^2\frac{P}{a^2} : (a'-\alpha)^2\frac{P'}{a'^2} : (a''-\alpha)^2\frac{P''}{a''^2} = (b-\beta)^2(b'-b'')^2 : (b'-\beta)^2(b''-b)^2 : (b''-\beta)^2(b-b')^2;$$

and thence

$$\begin{aligned} & (a-\alpha)\frac{\sqrt{(P)}}{a} : (a'-\alpha)\frac{\sqrt{(P')}}{a'} : (a''-\alpha)\frac{\sqrt{(P'')}}{a''} \\ &= (b-\beta)(b'-b'') : (b'-\beta)(b''-b) : (b''-\beta)(b-b'), \end{aligned}$$

which gives

$$(a-\alpha)\frac{\sqrt{(P)}}{a} + (a'-\alpha)\frac{\sqrt{(P')}}{a'} + (a''-\alpha)\frac{\sqrt{(P'')}}{a''} = 0,$$

and we have in like manner

$$(b-\beta)\frac{\sqrt{(P)}}{b} + (b'-\beta)\frac{\sqrt{(P')}}{b'} + (b''-\beta)\frac{\sqrt{(P'')}}{b''} = 0,$$

but the first of these equations is alone required for the present purpose. Putting for shortness

$$P=a^2X, \qquad P'=a'^2X', \qquad P''=a''^2X'',$$

the equation is

$$(a-\alpha)\sqrt{(X)}+(a'-\alpha)\sqrt{(X')}+(a''-\alpha)\sqrt{(X'')},$$

and by attributing the signs + and − to the radicals, we have, corresponding to the four conics, the equations

$$\begin{aligned}
(a-\alpha_1)\sqrt{(X)}+(a'-\alpha_1)\sqrt{(X')}+(a''-\alpha_1)\sqrt{(X'')}&=0,\\
-(a-\alpha_2)\sqrt{(X)}+(a'-\alpha_2)\sqrt{(X')}+(a''-\alpha_2)\sqrt{(X'')}&=0,\\
(a-\alpha_3)\sqrt{(X)}-(a'-\alpha_3)\sqrt{(X')}+(a''-\alpha_3)\sqrt{(X'')}&=0,\\
(a-\alpha_4)\sqrt{(X)}+(a'-\alpha_4)\sqrt{(X')}-(a''-\alpha_4)\sqrt{(X'')}&=0,
\end{aligned}$$

where α_1, α_2, α_3, α_4 are the values of α for the four conics respectively.

Eliminating a'' we obtain the system of three equations

$$\begin{aligned}
(2a-\alpha_1-\alpha_2)\sqrt{(X)}+(\alpha_2-\alpha_1)\sqrt{(X')}+(\alpha_2-\alpha_1)\sqrt{(X'')}&=0,\\
(\alpha_3-\alpha_1)\sqrt{(X)}+(2a'-\alpha_1-\alpha_3)\sqrt{(X')}+(\alpha_3-\alpha_1)\sqrt{(X'')}&=0,\\
(\alpha_1+\alpha_2-\alpha_3-\alpha_4)\sqrt{(X)}+(\alpha_1+\alpha_3-\alpha_2-\alpha_4)\sqrt{(X')}+(\alpha_1+\alpha_4-\alpha_2-\alpha_3)\sqrt{(X'')}&=0,
\end{aligned}$$

and then eliminating the radicals we have

$$\begin{vmatrix}
2a-\alpha_1-\alpha_2, & \alpha_2-\alpha_1, & \alpha_2-\alpha_1\\
\alpha_3-\alpha_1, & 2a'-\alpha_1-\alpha_3, & \alpha_3-\alpha_1\\
\alpha_1+\alpha_2-\alpha_3-\alpha_4, & \alpha_1+\alpha_3-\alpha_2-\alpha_4, & \alpha_1+\alpha_4-\alpha_2-\alpha_3
\end{vmatrix}=0,$$

which is in fact

$$-4\begin{vmatrix}
1, & a+a', & aa'\\
1, & \alpha_1+\alpha_4, & \alpha_1\alpha_4\\
1, & \alpha_2+\alpha_3, & \alpha_2\alpha_3
\end{vmatrix}=0,$$

as may be verified by actual expansion; the transformation of the determinant is a peculiar one.

The foregoing result was originally obtained as follows, viz. writing for a moment

$$\begin{aligned}
a\sqrt{(X)}+a'\sqrt{(X')}+a''\sqrt{(X'')}&=\Theta,\\
\sqrt{(X)}+\sqrt{(X')}+\sqrt{(X'')}&=\Phi,
\end{aligned}$$

the four equations are

$$\begin{aligned}
\Theta-\alpha_1\Phi&=0,\\
\Theta-\alpha_2\Phi&=2(a-\alpha_2)\sqrt{(X)},\\
\Theta-\alpha_3\Phi&=2(a'-\alpha_3)\sqrt{(X')},\\
\Theta-\alpha_4\Phi&=2(a''-\alpha_4)\sqrt{(X'')};
\end{aligned}$$

these give

$$\begin{aligned}
(\alpha_1-\alpha_2)\Phi&=2(a-\alpha_2)\sqrt{(X)},\\
(\alpha_1-\alpha_3)\Phi&=2(a'-\alpha_3)\sqrt{(X')},\\
(\alpha_1-\alpha_4)\Phi&=2(a''-\alpha_4)\sqrt{(X'')}.
\end{aligned}$$

From the last equation we have

$$(\alpha_1-\alpha_4)\,\Phi = 2\{\Theta - a\sqrt{(X)} - a'\sqrt{(X')}\} - 2\alpha_4\{\Phi - \sqrt{(X)} - \sqrt{(X')}\}$$
$$= 2(\alpha_1-\alpha_4)\,\Phi - 2(a-\alpha_4)\sqrt{(X)} - 2(a'-\alpha_4)\sqrt{(X')};$$

that is

$$(\alpha_1-\alpha_4)\,\Phi - 2(a-\alpha_4)\sqrt{(X)} - 2(a'-\alpha_4)\sqrt{(X')} = 0;$$

or substituting for $\sqrt{(X)}$, $\sqrt{(X')}$ their values in terms of Φ, we find

$$\alpha_1-\alpha_4 - \frac{(a-\alpha_4)(\alpha_1-\alpha_2)}{a-\alpha_2} - \frac{(a'-\alpha_4)(\alpha_1-\alpha_3)}{a'-\alpha_3} = 0,$$

which may be written

$$\alpha_1-\alpha_4-(\alpha_1-\alpha_2)\left(1+\frac{\alpha_2-\alpha_4}{a-\alpha_2}\right)-(\alpha_1-\alpha_3)\left(1+\frac{\alpha_3-\alpha_4}{a'-\alpha_3}\right)=0,$$

that is

$$\alpha_2+\alpha_3-\alpha_1-\alpha_4+\frac{(\alpha_2-\alpha_1)(\alpha_2-\alpha_4)}{a-\alpha_2}+\frac{(\alpha_3-\alpha_1)(\alpha_3-\alpha_4)}{a'-\alpha_3}=0;$$

or again

$$(\alpha_2-\alpha_1)\left(1+\frac{\alpha_2-\alpha_4}{a-\alpha_2}\right)+(\alpha_3-\alpha_4)\left(1+\frac{\alpha_3-\alpha_1}{a'-\alpha_3}\right)=0,$$

that is

$$(\alpha_2-\alpha_1)\frac{a-\alpha_4}{a-\alpha_2}+(\alpha_3-\alpha_4)\frac{a'-\alpha_1}{a'-\alpha_3}=0;$$

or finally

$$(\alpha_2-\alpha_1)(a-\alpha_4)(a'-\alpha_3)+(\alpha_3-\alpha_4)(a-\alpha_2)(a'-\alpha_1)=0,$$

which is a known form of the relation

$$\begin{vmatrix} 1, & a+a', & aa' \\ 1, & \alpha_1+\alpha_4, & \alpha_1\alpha_4 \\ 1, & \alpha_2+\alpha_3, & \alpha_2\alpha_3 \end{vmatrix} = 0,$$

which gives the involution of the quantities a, a'; α_1, α_4; α_2, α.

We have in like manner

$$\begin{vmatrix} 1, & a'+a'', & a'a'' \\ 1, & \alpha_1+\alpha_2, & \alpha_1\alpha_2 \\ 1, & \alpha_3+\alpha_4, & \alpha_3\alpha_4 \end{vmatrix} = 0,$$

and

$$\begin{vmatrix} 1, & a''+a, & a''a \\ 1, & \alpha_1+\alpha_3, & \alpha_1\alpha_3 \\ 1, & \alpha_2+\alpha_4, & \alpha_2\alpha_4 \end{vmatrix} = 0,$$

which give the involutions of the systems a'; a''; α_1, α_2; α_3, α_4 and a'', a; α_1, α_3; α_2, α_4 respectively.

It may be remarked that the equation of the conic passing through the three points and touching the axis of x in the point $x = \alpha$ is

$$\frac{(a-\alpha)^2(a'-a'')\,b}{bx+ay-ab}+\frac{(a'-\alpha)^2(a''-a)\,b'}{b'x+a'y-a'b'}+\frac{(a''-\alpha)^2(a-a')\,b''}{b''x+a''y-a''b''}=0,$$

and when this meets the axis of y we have

$$\frac{\frac{b}{a}(a-\alpha)^2(a'-a'')}{y-b}+\frac{\frac{b'}{a'}(a'-\alpha)^2(a''-a)}{y-b'}+\frac{\frac{b''}{a''}(a''-\alpha)^2(a-a')}{y-b''}=0.$$

Hence, if this touches the axis of y in the point $y=\beta$, the left-hand side must be

$$=\frac{\left[\frac{b}{a}(a-\alpha)^2(a'-a'')+\frac{b'}{a'}(a'-\alpha)^2(a''-a)+\frac{b''}{a''}(a''-\alpha)^2(a-a')\right](y-\beta)^2}{(y-b)(y-b')(y-b'')},$$

and equating the coefficients of $\frac{1}{y^2}$, we have

$$\frac{b^2}{a}(a-\alpha)^2(a'-a'')+\frac{b'^2}{a'}(a'-\alpha)^2(a''-a)+\frac{b''^2}{a''}(a''-\alpha)^2(a-a')$$

$$=\left[\frac{b}{a}(a-\alpha)^2(a'-a'')+\frac{b'}{a'}(a'-\alpha)^2(a''-a)+\frac{b''}{a''}(a''-\alpha)^2(a-a')\right](b+b'+b''-2\beta),$$

or what is the same thing,

$$\frac{b(b'+b'')}{a}(a-\alpha)^2(a'-a'')+\frac{b'(b''+b)}{a'}(a'-\alpha)^2(a''-a)+\frac{b''(b+b')}{a''}(a''-\alpha)^2(a-a')$$

$$=2\beta\left[\frac{b}{a}(a-\alpha)^2(a'-a'')+\frac{b'}{a'}(a'-\alpha)^2(a''-a)+\frac{b''}{a''}(a''-\alpha)^2(a-a')\right],$$

which gives β in terms of α, that is β_1, β_2, β_3, β_4 in terms of α_1, α_2, α_3, α_4 respectively.

Cambridge, 30 November, 1863.

391.

SOLUTION OF A PROBLEM OF ELIMINATION.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. VIII. (1867), pp. 183—185.]

It is required to eliminate x, y from the equations

$$\left\|\begin{matrix} x^4, & x^3y, & x^2y^2, & xy^3, & y^4 \\ a, & b, & c, & d, & e \\ a', & b', & c', & d', & e' \\ a'', & b'', & c'', & d'', & e'' \end{matrix}\right\| = 0.$$

This system may be written

$$\begin{aligned} x^4 &= \Sigma\lambda a, \\ x^3y &= \Sigma\lambda b, \\ x^2y^2 &= \Sigma\lambda c, \\ xy^3 &= \Sigma\lambda d, \\ y^4 &= \Sigma\lambda e; \end{aligned}$$

if for shortness

$$\Sigma\lambda a = \lambda a + \lambda' a' + \lambda'' a'', \text{ \&c.}$$

Or putting

$$\frac{x}{y} = -k,$$

we have

$$\begin{aligned} \Sigma\lambda\,(a + kb) &= 0, \\ \Sigma\lambda\,(b + kc) &= 0, \\ \Sigma\lambda\,(c + kd) &= 0, \\ \Sigma\lambda\,(d + ke) &= 0; \end{aligned}$$

or, what is the same thing,

$$\begin{aligned}
&\lambda (a+kb)+\lambda' (a'+kb')+\lambda'' (a''+kb'')=0,\\
&\lambda (b+kc)+\lambda' (b'+kc')+\lambda'' (b''+kc'')=0,\\
&\lambda (c+kd)+\lambda' (c'+kd')+\lambda'' (c''+kd'')=0,\\
&\lambda (d+ke)+\lambda' (d'+ke')+\lambda'' (d''+ke'')=0\,;
\end{aligned}$$

and representing the columns

$$\begin{matrix}
a & b, & a' & b', & a'' & b'',\\
b & c, & b' & c', & b'' & c'',\\
c & d, & c' & d', & c'' & d'',\\
d & e, & d' & e', & d'' & e'',
\end{matrix}$$

by

$$1,\ 2,\ \ 3,\ 4,\ \ 5,\ 6,$$

each equation is of the type

$$\lambda (1+k2)+\lambda' (3+k4)+\lambda'' (5+k6)=0.$$

Multiplying the several equations by the minors of 135, each with its proper sign, and adding, the terms independent of k disappear, the equation divides by k, and we find

$$\lambda\, 2135+\lambda'\, 4135+\lambda''\, 6135=0\,;$$

operating in a similar manner with the minors of 246, the terms in k disappear, and we find

$$\lambda\, 1246+\lambda'\, 3246+\lambda''\, 5246=0\,;$$

again, operating with the minors of $(146+236+245+k246)$, we find

$$\begin{aligned}
&\lambda\ \{1236+1245+k(2146+1246)\}\\
&+\lambda'\ \{3146+3245+k(4236+3246)\}\\
&+\lambda''\ \{5146+5236+k(6245+5246)\}=0,
\end{aligned}$$

where the terms in k disappear, and this is

$$\lambda (1236+1245)+\lambda' (3146+3245)+\lambda'' (5146+5236)=0.$$

We have thus three linear equations, which written in a slightly different form are

$$\begin{aligned}
&\lambda\ 1235 &&+\lambda'\ 3451 &&+\lambda''\ 5613 &&=0,\\
&\lambda\ (1236+1245) &&+\lambda'\ (3452+3461) &&+\lambda''\ (5614+5623) &&=0,\\
&\lambda\ 1246 &&+\lambda'\ 3462 &&+\lambda''\ 5624 &&=0,
\end{aligned}$$

and thence eliminating λ, λ', λ'', we have

$$\begin{vmatrix}
1235, & 1236+1245, & 1246\\
3451, & 3452+3461, & 3462\\
5613, & 5614+5623, & 5624
\end{vmatrix}=0,$$

which is the required result. It may be remarked that the second and third column are obtained from the first by operating on it with Δ, $\frac{1}{2}\Delta^2$, if $\Delta = 2\delta_1 + 4\delta_3 + 6\delta_5$. Or say the result is

$$(1,\ \Delta,\ \tfrac{1}{2}\Delta^2)\ \left|\begin{matrix} 1235 \\ 3451 \\ 5613 \end{matrix}\right| = 0.$$

In like manner for the system

$$\left\|\begin{matrix} x^5\ , & x^4y, & x^3y^2, & x^2y^3, & xy^4, & y^5 \\ a\ , & b\ , & c\ , & d\ , & e\ , & f \\ a'\ , & b'\ , & c'\ , & d'\ , & e'\ , & f' \\ a''\ , & b''\ , & c''\ , & d''\ , & e''\ , & f'' \\ a''', & b''', & c'''\ , & d''', & e'''\ , & f''' \end{matrix}\right\| = 0,$$

if the columns are

$$\begin{matrix} a\ b, & a'\ b', & a''\ b'', & a'''\ b''', \\ b\ c, & b'\ c', & b''\ c'', & b'''\ c''', \\ c\ d, & c'\ d', & c''\ d'', & c'''\ d''', \\ d\ e, & d'\ e', & d''\ e'', & d'''\ e''', \\ e\ f, & e'\ f', & e''\ f'', & e'''\ f''', \\ =1,\ 2, & 3,\ 4, & 5,\ 6, & 7,\ 8\,; \end{matrix}$$

then the result is

$$(1,\ \Delta,\ \tfrac{1}{2}\Delta^2,\ \tfrac{1}{6}\Delta^3)\ \left|\begin{matrix} 12357 \\ 34571 \\ 56713 \\ 78135 \end{matrix}\right| = 0,$$

where

$$\Delta = 2\delta_1 + 4\delta_3 + 6\delta_5 + 8\lambda_7.$$

392.

ON THE CONICS WHICH PASS THROUGH TWO GIVEN POINTS AND TOUCH TWO GIVEN LINES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. VIII. (1867), pp. 211—219.]

LET $x=0$, $y=0$ be the equations of the given lines; $z=0$ the equation of the line joining the given points. We may, to fix the ideas, imagine the implicit constants so determined that $x+y+z=0$ shall be the equation of the line infinity.

Take $x-my=0$, $x-ny=0$ as the equations of the lines which by their intersection with $z=0$ determine the given points. The equation of the conic is

$$\{\sqrt{(m)}+\sqrt{(n)}\}\sqrt{(xy)}=x+y\sqrt{(mn)}+\gamma z,$$

or, what is the same thing,

$$(x-my)(x-ny)+2\{x+y\sqrt{(mn)}\}\gamma z+\gamma^2z^2=0,$$

so that there are two distinct series of conics according as $\sqrt{(mn)}$ is taken with the positive or the negative sign.

The equation of the chord of contact is

$$x+y\sqrt{(mn)}+\gamma z=0,$$

which meets $z=0$ in the point $\{x+y\sqrt{(mn)}=0,\ z=0\}$ that is in one of the centres of the involution formed by the lines $(x=0,\ y=0)$, $(x-my=0,\ x-ny=0)$. It is to be observed that the conic is only real when mn is positive, that is (the lines and points being each real) the two points must be situate in the same region or in opposite regions of the four regions formed by the two lines: there are however other real cases; e.g. if the lines $x=0$, $y=0$ are real, but the quantities m, n are conjugate imaginaries; included in this we have the circles which touch two real lines.

To fix the ideas I take m and n each positive and $mn > 1$; also I attend first to the series where $\sqrt{(mn)}$ is taken positively. At the points where the conic meets infinity, we have

$$\{\sqrt{(m)} + \sqrt{(n)}\}\sqrt{(xy)} = x + y\sqrt{(mn)} - \gamma(x+y),$$

which gives two coincident points, that is the conic is a parabola, if

$$(1-\gamma)\{\sqrt{(mn)} - \gamma\} = \tfrac{1}{4}\{\sqrt{(m)} + \sqrt{(n)}\}^2,$$

that is

$$\gamma^2 - \gamma\{1 + \sqrt{(mn)}\} = \tfrac{1}{4}\{\sqrt{(m)} - \sqrt{(n)}\}^2,$$

or

$$\gamma = \tfrac{1}{2}[1 + \sqrt{(mn)} \pm \sqrt{\{(1+m)(1+n)\}}],$$

where it is to be noticed that

$$\gamma = \tfrac{1}{2}[1 + \sqrt{(mn)} + \sqrt{\{(1+m)(1+n)\}}]$$

is a positive quantity greater than $\sqrt{(mn)}$, say $\gamma = p$,

$$\gamma = \tfrac{1}{2}[1 + \sqrt{(mn)} - \sqrt{\{(1+m)(1+n)\}}]$$

is a negative quantity, say $\gamma = -q$, q being positive.

The order of the lines is as shown in fig. 1, see plate facing p. 52.

$\gamma = -\infty$ to $\gamma = -q$, curve is ellipse; $\gamma = -q$, parabola P_2,
$\gamma = -q$ to p, curve is hyperbola; $\gamma = p$, parabola P_1,
$\gamma = p$ to $\gamma = \infty$, ellipse.

Resuming the equation

$$(x - my)(x - ny) + 2\{x + y\sqrt{(mn)}\}\gamma z + \gamma^2 z^2 = 0,$$

the coefficients are

$$(a, b, c, f, g, h) = \{1,\ mn,\ \gamma^2,\ \gamma\sqrt{(mn)},\ \gamma,\ -\tfrac{1}{2}(m+n)\},$$

and thence the inverse coefficients are

$$(A, B, C, F, G, H) =$$

$$[0,\ 0,\ -\tfrac{1}{4}(m-n)^2,\ -\tfrac{1}{2}\gamma\{\sqrt{(m)}+\sqrt{(n)}\}^2,\ -\tfrac{1}{2}\gamma\sqrt{(mn)}\{\sqrt{(m)}+\sqrt{(n)}\}^2,\ \tfrac{1}{2}\gamma^2\{\sqrt{(m)}+\sqrt{(n)}\}^2],$$

$$K = -\tfrac{1}{4}\gamma^2\{\sqrt{(m)} + \sqrt{(n)}\}^4,$$

or, omitting a factor, the inverse coefficients are

$$(A, B, C, F, G, H) = \left[0,\ 0,\ \frac{1}{2\gamma}\{\sqrt{(m)} - \sqrt{(n)}\}^2,\ 1,\ \sqrt{(mn)},\ -\gamma\right].$$

Considering the line

$$\lambda x + \mu y + \nu z = 0,$$

the coordinates of the pole of this line are

$$\begin{aligned} x : y : z = &\quad -\gamma\mu + \sqrt{(mn)}\,\nu \\ &: -\gamma\lambda + \nu \\ &: \sqrt{(mn)}\,\lambda + \mu + \frac{1}{2\gamma}\{\sqrt{(m)} - \sqrt{(n)}\}^2\nu, \end{aligned}$$

or (what is the same thing) introducing the arbitrary coefficient k, we have

$$\begin{aligned} &kx + \gamma\mu - \nu\sqrt{(mn)} = 0, \\ &ky + \gamma\lambda - \nu \qquad\quad = 0, \\ &kz - \lambda\sqrt{(mn)} - \mu - \frac{1}{2\gamma}\{\sqrt{(m)} - \sqrt{(n)}\}^2 \nu = 0\,; \end{aligned}$$

the first two equations give

$$k : \gamma : -1 = \nu\{\mu - \lambda\sqrt{(mn)}\} : \nu\{y\sqrt{(mn)} - x\} : \lambda x - \mu y,$$

that is

$$k = \frac{-\nu\{\mu - \lambda\sqrt{(mn)}\}}{\lambda x - \mu y}, \quad \gamma = \frac{-\nu\{y\sqrt{(mn)} - x\}}{\lambda x - \mu y},$$

or, substituting this value of γ in the third equation,

$$\frac{\nu\{\mu - \lambda\sqrt{(mn)}\}\, z}{\lambda x - \mu y} + \{\mu + \lambda\sqrt{(mn)}\} + \frac{\lambda x - \mu y}{x - y\sqrt{(mn)}}\, \frac{\{\sqrt{(m)} - \sqrt{(n)}\}^2}{2} = 0,$$

that is

$$(\lambda x - \mu y)^2 . \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 + \{x - y\sqrt{(mn)}\}(\lambda x - \mu y)\{\mu + \lambda\sqrt{(mn)}\} + z\{x - y\sqrt{(mn)}\}\,\nu\{\mu - \lambda\sqrt{(mn)}\} = 0,$$

which is the equation of the curve, the locus of the pole of the line $\lambda x + \mu y + \nu z = 0$ in regard to the conic

$$(x - my)(x - ny) + 2\{x + y\sqrt{(mn)}\}\gamma z + \gamma^2 z^2 = 0.$$

In particular, if $\lambda = \mu = \nu = 1$, then for the coordinates of the centre of the conic, we have

$$x : y : z = -\gamma + \sqrt{(mn)} : -\gamma + 1 : \sqrt{(mn)} + 1 + \frac{1}{2\gamma}\{\sqrt{(m)} - \sqrt{(n)}\}^2;$$

and for the locus of the centre,

$$(x - y)^2 . \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 + (x - y)\{x - y\sqrt{(mn)}\}\{1 + \sqrt{(mn)}\} + z\{x - y\sqrt{(mn)}\}\{1 - \sqrt{(mn)}\} = 0,$$

so that the locus is a conic, and it is obvious that this conic is a hyperbola. Putting for greater simplicity

$$\begin{aligned} x - y \qquad &= X, \\ x - y\sqrt{(mn)} &= Y, \\ z \qquad &= Z, \end{aligned}$$

the equation of the curve of centres is

$$X^2 . \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 + XY\{1 + \sqrt{(mn)}\} + YZ\{1 - \sqrt{(mn)}\} = 0,$$

or, writing this under the form

$$Y[X\{1 + \sqrt{(mn)}\} + Z\{1 - \sqrt{(mn)}\}] + \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 X^2 = 0,$$

the equation is

$$YQ + X^2 = 0,$$

where

$$X = x - y,$$
$$Y = x - y\sqrt{(mn)},$$
$$Q = \frac{2}{\{\sqrt{(m)} - \sqrt{(n)}\}^2}\left[\{1 + \sqrt{(mn)}\}(x - y) + \{1 - \sqrt{(mn)}\}z\right]:$$

these values give

$$x - y \qquad = X,$$
$$x - y\sqrt{(mn)} = Y,$$
$$\{1 - \sqrt{(mn)}\}z = \{\sqrt{(m)} - \sqrt{(n)}\}^2 Q + 2\{1 + \sqrt{(mn)}\}X,$$

or, what is the same thing,

$$\{1 - \sqrt{(mn)}\}x = \quad -\sqrt{(mn)}\; X + Y,$$
$$\{1 - \sqrt{(mn)}\}y = \qquad\qquad - X + Y,$$
$$\{1 - \sqrt{(mn)}\}z = 2\{1 + \sqrt{(mn)}\}X \qquad + \{\sqrt{(m)} - \sqrt{(n)}\}^2 Q,$$

whence also

$$\{1 - \sqrt{(mn)}\}(x + y + z) = \{1 + \sqrt{(mn)}\}X + 2Y + \{\sqrt{(m)} - \sqrt{(n)}\}^2 Q,$$

or the equation of the line infinity is

$$\{1 + \sqrt{(mn)}\}X + 2Y + \{\sqrt{(m)} - \sqrt{(n)}\}^2 Q = 0,$$

a formula which may be applied to finding the asymptotes and thence the centre of the conic

$$YQ + X^2 = 0.$$

In fact we have identically

$$\{2kx + 2ky - (2k+1)z\}^2 - (1 + 4k)(2kx - z)^2 = 4k^2(x + y + z)^2 - 4k(1 + 4k)(kx^2 + yz),$$

that is

$$-4k(1 + 4k)(kx^2 + yz) = \{2kx + 2ky - (2k+1)z\}^2 - (1 + 4k)(2kx - z)^2 - 4k^2(x + y + z)^2,$$

which, if $x + y + z = 0$ is the equation of the line infinity, puts in evidence the asymptotes of the conic $kx^2 + yz = 0$. Hence writing αx, βy, γz in the place of x, y, z respectively, and $\frac{k\alpha^2}{\beta\gamma} = k'$, that is, $k = \frac{\beta\gamma}{\alpha^2}k'$, we have

$$-\frac{4\beta\gamma}{\alpha^2}k'\left(1 + \frac{4\beta\gamma}{\alpha^2}k'\right)\beta\gamma(k'x^2 + yz) = \left\{\frac{2\beta\gamma}{\alpha}k'x + \frac{2\beta^2\gamma}{\alpha^2}k'y - \left(2\frac{\beta\gamma}{\alpha^2}k' + 1\right)\gamma z\right\}^2$$
$$- \left(1 + \frac{4\beta\gamma}{\alpha^2}k'\right)\left(\frac{2\beta\gamma}{\alpha}k'x - \gamma z\right)^2 - 4\frac{\beta^2\gamma^2}{\alpha^4}k'^2(\alpha x + \beta y + \gamma z)^2,$$

that is

$$-4\beta^2\gamma^2k'(\alpha^2+4\beta\gamma k')(k'x^2+yz)=\{2\alpha\beta\gamma k'x+2\beta^2\gamma k'y-(2\beta\gamma k'+\alpha^2)\gamma z\}^2$$
$$-(\alpha^2+4\beta\gamma k')(2\beta\gamma k'x-\alpha\gamma z)^2-4\beta^2\gamma^2k'^2(\alpha x+\beta y+\gamma z)^2,$$

or, what is the same thing,

$$-4\beta^2k'(\alpha^2+4\beta\gamma k')(k'x^2+yz)=\{2\alpha\beta k'x+2\beta^2k'y-(2\beta\gamma k'+\alpha^2)z\}^2$$
$$-(\alpha^2+4\beta\gamma k')(2\beta k'x-\alpha z)^2-4\beta^2k'^2(\alpha x+\beta y+\gamma z)^2,$$

which, when $\alpha x+\beta y+\gamma z=0$ is the equation of the line infinity, puts in evidence the asymptotes of the conic $k'x^2+yz=0$.

Now writing X, Y, Q in the place of x, y, z; $k'=1$, and $\alpha=\{1+\surd(mn)\}$, $\beta=2$, $\gamma=\{\surd(m)-\surd(n)\}^2$, we have

$$-16[\{1+\surd(mn)\}^2+8\{\surd(m)-\surd(n)\}^2](YQ+X^2)$$
$$=\ [4\{1+\surd(mn)\}X+8Y-(4\{\surd(m)-\surd(n)\}^2+\{1+\surd(mn)\}^2)Q]^2$$
$$-[\{1+\surd(mn)\}^2+8\{\surd(m)-\surd(n)\}^2][4X-\{1+\surd(mn)\}Q]^2$$
$$-16[\{1+\surd(mn)\}X+2Y+\{\surd(m)-\surd(n)\}^2Q]^2,$$

and the asymptotes are

$$4\{1+\surd(mn)\}X+8Y-[4\{\surd(m)-\surd(n)\}^2+\{1+\surd(mn)\}^2]Q$$
$$=\pm\sqrt{\{1+\surd(mn)\}^2+8\{\surd(m)-\surd(n)\}^2}\,[4X-\{1+\surd(mn)\}Q].$$

At the centre

$$4\{1+\surd(mn)\}X+8Y-[4\{\surd(m)-\surd(n)\}^2+\{1+\surd(mn)\}^2]Q=0,$$
$$4X-\{1+\surd(mn)\}Q=0,$$

but the first equation is

$$\{1+\surd(mn)\}[4X-Q\{1+\surd(mn)\}]+8Y-4\{\surd(m)-\surd(n)\}^2Q=0,$$

so that we have

$$4X=\{1+\surd(mn)\}Q,\quad 2Y=\{\surd(m)-\surd(n)\}^2Q,$$

the first of these is

$$2\{\surd(m)-\surd(n)\}^2(x-y)-\{1+\surd(mn)\}^2(x-y)-(1-mn)z=0,$$

and the two together give

$$2X\{\surd(m)-\surd(n)\}^2-\{1+\surd(mn)\}Y=0,$$

so that we have

$$2\{\surd(m)-\surd(n)\}^2(x-y)-\{1+\surd(mn)\}\{x-y\surd(mn)\}=0,$$
$$[\{1+\surd(mn)\}^2-2\{\surd(m)-\surd(n)\}^2](x-y)+(1-mn)z=0,$$

to determine the coordinates of the centre.

The equation of the chord of contact is

$$x + y\sqrt{(mn)} + \gamma z = 0,$$

which for $\gamma = 1$ is parallel to $y = 0$ and for $\gamma = \sqrt{(mn)}$ is parallel to $x = 0$. But the coordinates of the centre are

$$x : y : z = -\gamma + \sqrt{(mn)} : -\gamma + 1 : \sqrt{(mn)} + 1 + \frac{1}{2\gamma}\{\sqrt{(m)} - \sqrt{(n)}\}^2,$$

which for $\gamma = 1$ give

$$y = 0,\ x : z = -1 + \sqrt{(mn)} : \sqrt{(mn)} + 1 + \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2, = -2 + 2\sqrt{(mn)} : 2 + m + n,$$

and for $\gamma = \sqrt{(mn)}$ give

$$x = 0,$$

$$y : z = 1 - \sqrt{(mn)} : \sqrt{(mn)} + 1 + \frac{1}{2\sqrt{(mn)}}\{\sqrt{(m)} - \sqrt{(n)}\}^2,\ = 2 - 2\sqrt{(mn)} : 2\sqrt{(mn)} + \frac{m+n}{\sqrt{(mn)}}.$$

The line drawn from the fixed point on the chord of contact to the centre has for its equation

$$x + y\sqrt{(mn)} + \gamma' z = 0,$$

where, writing for x, y, z the coordinates of the centre, we have

$$-\gamma\{1 + \sqrt{(mn)}\} + 2\sqrt{(mn)} + \gamma'\left[\sqrt{(mn)} + 1 + \frac{1}{2\gamma}\{\sqrt{(m)} - \sqrt{(n)}\}^2\right] = 0,$$

that is

$$\gamma' = \frac{\gamma\{1 + \sqrt{(mn)}\} - 2\sqrt{(mn)}}{1 + \sqrt{(mn)} + \frac{1}{2\gamma}\{\sqrt{(m)} - \sqrt{(n)}\}^2},$$

or, what is the same thing,

$$\gamma' - \gamma = \frac{-\gamma\{\sqrt{(m)} + \sqrt{(n)}\}^2}{\{\sqrt{(m)} - \sqrt{(n)}\}^2 + 2\gamma\{1 + \sqrt{(mn)}\}},$$

and consequently $\gamma' = \gamma$ only for $\gamma = 0$.

It is now easy to trace the corresponding positions of the chord of contact through the fixed point $\{x + y\sqrt{(mn)} = 0,\ z = 0\}$, and of the centre on the hyperbola which is the curve of centres: see fig. 2 in the plate facing p. 52.

The lines OP_2, OL, $O\Theta$, OP_1, OX, OG, OH are positions of the chord of contact, and the points P_2, L, Θ, P_1, X, G, H, on the hyperbola which is the curve of centres are the corresponding positions of the centre.

Chord of Contact.	*Centre.*
OP_2.	P_2, at infinity on hyperbola.
$OL\,(z=0)$.	L, $(z=0,\ x-y=0)$.
$O\Theta$.	Θ, the line joining this with O being always behind $O\Theta$.
OP_1.	P_1, at infinity on hyperbola.
$OX\ \{x+y\sqrt{(mn)}=0\}$.	$X\ (x=0,\ y=0)$.
OG (parallel to $y=0$).	G (on line $y=0$).
OH (parallel to $x=0$) and so back to	H (on line $x=0$) and so on to
OP_2.	P_2.

I have treated separately the case $\sqrt{(mn)}=1$.

Consider the conics which touch the lines $y-x=0$, $y+x=0$ and pass through the points

$$\{x=1,\ y=\sqrt{(1-c^2)}\},\ \{x=1,\ y=-\sqrt{(1-c^2)}\}.$$

The equation is of the form

$$y^2-x^2+k(x-\alpha)^2=0,$$

and to determine k, we have

$$1-c^2-1+k(1-\alpha)^2=0, \text{ and therefore } k=\frac{c^2}{(1-\alpha)^2}.$$

The equation thus becomes

$$(1-\alpha)^2(y^2-x^2)+c^2(x-\alpha)^2=0,$$

that is

$$(1-\alpha)^2y^2+\{c^2-(1-\alpha)^2\}x^2-2c^2\alpha x+c^2\alpha^2=0,$$

or as this may be written

$$(1-\alpha)^2y^2+\{c^2-(1-\alpha)^2\}\left\{x-\frac{c^2\alpha}{c^2-(1-\alpha)^2}\right\}^2-\frac{c^2\alpha^2(1-\alpha)^2}{c^2-(1-\alpha)^2}=0.$$

Hence the nature of the conic depends on the sign of $c^2-(1-\alpha)^2$, viz. if this be positive, or α between the limits $1+c$, $1-c$, the curve is an ellipse,

$$x\text{-coordinate of centre}=\frac{c^2\alpha}{c^2-(1-\alpha)^2},$$

which is positive,

$$x\text{-semi-axis}=\frac{\pm c\alpha(1-\alpha)}{c^2-(1-\alpha)^2},$$

$$y\text{-semi-axis}=\frac{c\alpha}{\sqrt{\{c^2-(1-\alpha)^2\}}}.$$

The coordinate of centre for $\alpha=1+c$ is $=+\infty$ (the curve being in this case a parabola P_1) and for $\alpha=1-c$ it is also $=+\infty$ (the curve being in this case a parabola P_2). The coordinate has a minimum value corresponding to $\alpha=\sqrt{(1-c^2)}$, viz. this is $=\frac{1}{2}\{1+\sqrt{(1-c^2)}\}$.

Hence as (α) passes from $1+c$ to $\sqrt{(1-c^2)}$, the coordinate of the centre passes from ∞ to its minimum value $\frac{1}{2}\{1+\sqrt{(1-c^2)}\}$; in the passage we have $\alpha=1$ giving the coordinate $=1$, the conic being in this case a pair of coincident lines $(x-1)^2=0$. And as (α) passes from the foregoing value $\sqrt{(1-c^2)}$ to $1-c$, the coordinate of the centre passes from the minimum value $\frac{1}{2}\{1+\sqrt{(1-c^2)}\}$ to ∞.

The curve is a hyperbola if α lies without the limits $1+c$, $1-c$,

$$x\text{-coordinate of centre} \quad = \frac{-c^2\alpha}{(1-\alpha)^2-c^2},$$

which has the sign of $-\alpha$,

$$x\text{-semi-axis} \quad = \frac{\pm c\alpha(1-\alpha)}{(1-\alpha)^2-c^2},$$

$$y\text{-semi-axis} \quad = \frac{\pm c\alpha}{\sqrt{\{(1-\alpha)^2-c^2\}}},$$

$$\text{semi-aperture of asymptotes} = \tan^{-1}\sqrt{\left\{1-\frac{c^2}{(1-\alpha)^2}\right\}},$$

which for $\alpha=1\pm c$ is $=0$ (parabola), but increases as $1-\alpha$ increases positively or negatively, becoming $=45°$ for $\alpha=\pm\infty$ (the asymptotes being in this case the pair of lines $y^2-x^2=0$):

$$\alpha=+\infty,\ \text{coordinate of centre is } = 0,$$

$$\alpha=1+c,\quad \text{,,} \quad \text{,,} \quad =-\infty,$$

so that α diminishing from ∞ to $1+c$, the coordinate of the centre moves constantly in the same direction from 0 to $-\infty$,

$$\alpha=1-c,\ \text{coordinate of centre is } =-\infty,$$

$$\alpha=0,\quad \text{,,} \quad \text{,,} \quad = 0,$$

the hyperbola being in this case the pair of lines $y^2=(1-c^2)x^2$.

α negative, the coordinate of centre becomes positive, viz. as α passes from $\alpha=0$ to $\alpha=-\sqrt{(1-c^2)}$, the coordinate of centre passes from 0 to a maximum positive value $\frac{1}{2}\{1-\sqrt{(1-c^2)}\}$, and then as α passes from $-\sqrt{(1-c^2)}$ to $-\infty$, the coordinate of centre diminishes from $\frac{1}{2}\{1-\sqrt{(1-c^2)}\}$ to 0. It is to be remarked that α being negative, the lines $y^2-x^2=0$ are touched by the branch on the negative side of the origin, that is the branch not passing through the two points $x=1$, $y=\pm\sqrt{(1-c^2)}$.

393.

ON THE CONICS WHICH TOUCH THREE GIVEN LINES AND PASS THROUGH A GIVEN POINT.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. VIII. (1867), pp. 220—222.]

CONSIDER the triangles which touch three given lines; the three lines form a triangle, and the lines joining the angles of the triangle with the points of contact of the opposite sides respectively meet in a point S: conversely given the three lines and the point S, then joining this point with the angles of the triangle the joining lines meet the opposite sides respectively in three points which are the points of contact with the three given lines respectively of a conic; such conic is determinate and unique. Suppose now that the conic passes through a given point; the point S is no longer arbitrary, but it must lie on a certain curve; and this curve being known, then taking upon it any point whatever for the point S, and constructing as before the conic which corresponds to such point, the conic in question will pass through the given point, and will thus be a conic touching the three given lines and passing through the given point. And the series of such conics corresponds of course to the series of points on the curve.

I proceed to find the curve which is the locus of the point S.

We may take $x=0$, $y=0$, $z=0$ for the equations of the given lines, and $x : y : z = 1 : 1 : 1$ for the coordinates of the given point. The equation of a conic touching the three given lines is

$$a\sqrt{(x)} + b\sqrt{(y)} + c\sqrt{(z)} = 0,$$

and the coordinates of the corresponding point S are as $\frac{1}{a^2} : \frac{1}{b^2} : \frac{1}{c^2}$, that is, taking (x, y, z) for the coordinates of the point in question, we have

$$a : b : c = \frac{1}{\sqrt{(x)}} : \frac{1}{\sqrt{(y)}} : \frac{1}{\sqrt{(z)}},$$

the condition in order that the conic may pass through the given point is $a+b+c=0$, and we thus find for the curve, which is the locus of the point S, the equation

$$\frac{1}{\sqrt{(x)}}+\frac{1}{\sqrt{(y)}}+\frac{1}{\sqrt{(z)}}=0,$$

or, what is the same thing,

$$\sqrt{(yz)}+\sqrt{(zx)}+\sqrt{(xy)}=0,$$

the rationalised form of which is

$$y^2z^2+z^2x^2+x^2y^2-2xyz(x+y+z)=0.$$

This is a quartic curve with three cusps, viz. each angle of the triangle is a cusp; and by considering for example the cusp ($y=0$, $z=0$) and writing the equation under the form

$$x^2(y-z)^2-2x(yz^2+y^2z)+y^2z^2=0,$$

we see that the tangent at the cusp in question is the line $y-z=0$; that is, the tangents at the three cusps are the lines joining these points respectively with the given point (1, 1, 1). Each cuspidal tangent meets the curve in the cusp counting as three points and in a fourth point of intersection, the coordinates whereof in the case of the tangent $y-z=0$, are at once found to be $x:y:z=1:4:4$, or say this is the point (1, 4, 4); the point on the tangent $z-x=0$ is of course (4, 1, 4), and that on the tangent $x-y=0$ is (4, 4, 1). To find the tangents at these points respectively, I remark that the general equation of the tangent is

$$(X\delta_x+Y\delta_y+Z\delta_z)\left\{\frac{1}{\sqrt{(x)}}+\frac{1}{\sqrt{(y)}}+\frac{1}{\sqrt{(z)}}\right\}=0,$$

that is

$$\frac{X}{x^{\frac{3}{2}}}+\frac{Y}{y^{\frac{3}{2}}}+\frac{Z}{z^{\frac{3}{2}}}=0,$$

or for the point (1, 4, 4) the equation of the tangent is $8X+Y+Z=0$, or say $8x+y+z=0$; that is, the tangent passes through the point $x=0$, $x+y+z=0$, being the point of intersection of the line $x=0$ with the line $x+y+z=0$, which is the harmonic of the given point (1, 1, 1) in regard to the triangle; the tangents at the points (1, 4, 4), (4, 1, 4), (4, 4, 1) respectively pass through the points of intersection of the harmonic line $x+y+z=0$ with the three given lines respectively.

In the case where the given point lies within the triangle, the curve the locus of S lies wholly within the triangle, and is of the form shown in fig. 3 in the plate opposite; it is clear that in this case the conics of the system are all of them ellipses; there are however three limiting forms, viz. the line joining the given point with any angle of the triangle, such line being regarded as a twofold line or pair of coincident lines, is a conic of the system. The discussion of the two cases in which the given point lies outside the triangle, viz. in the infinite space bounded by two sides produced, or in the infinite space bounded by a side and two sides produced, may be effected without much difficulty.

Plate I.

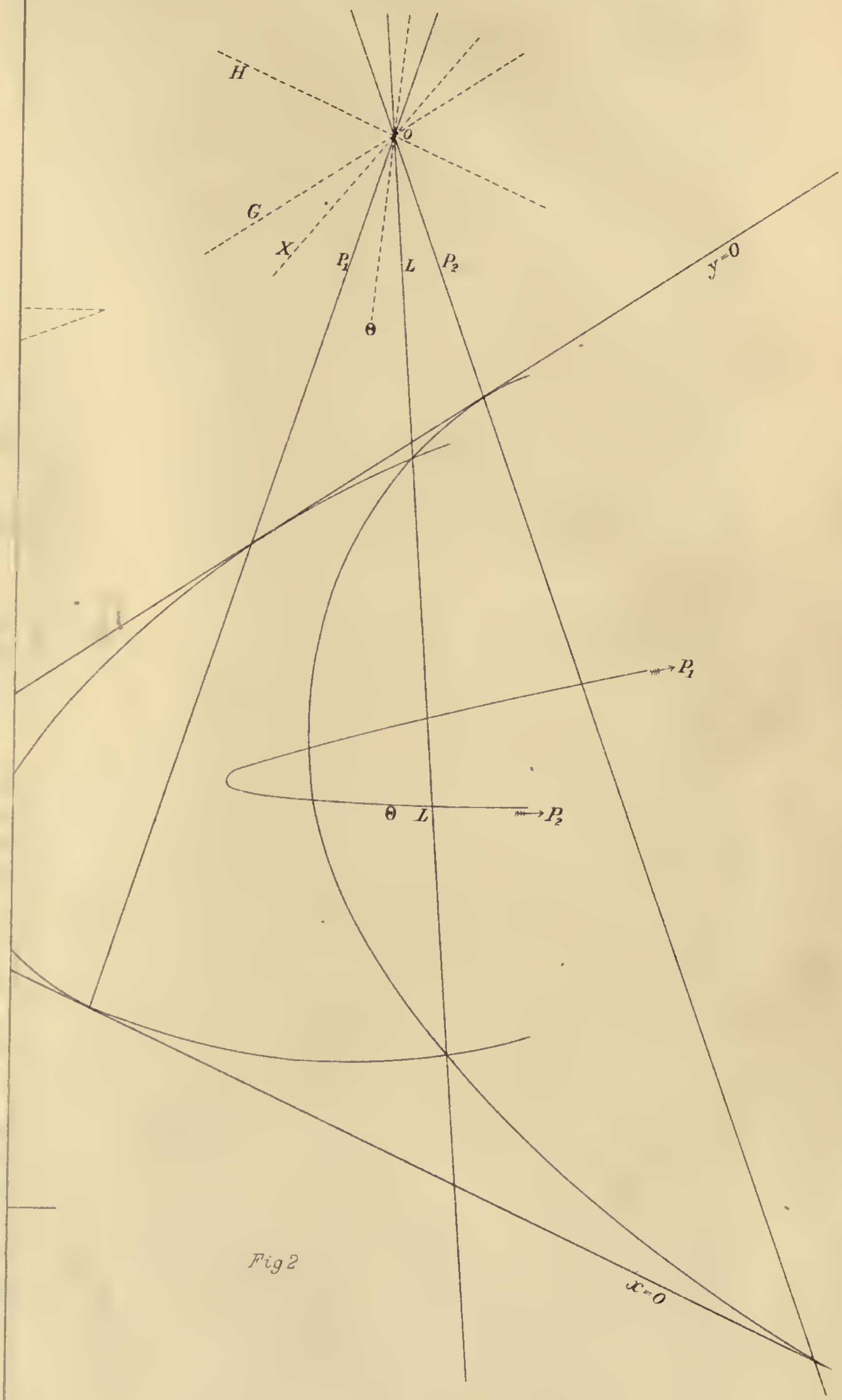


Fig 2

To face p. 52.

394.

ON A LOCUS IN RELATION TO THE TRIANGLE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. VIII. (1867), pp. 264—277.]

IF from any point of a circle circumscribed about a triangle perpendiculars are let fall upon the sides, the feet of the perpendiculars lie in a line; or, what is the same thing, the locus of a point, such that the perpendiculars let fall therefrom upon the sides of a given triangle have their feet in a line, is the circle circumscribed about the triangle.

In this well known theorem we may of course replace the circular points at infinity by any two points whatever; or the Absolute being a point-pair, and the terms perpendicular and circle being understood accordingly, we have the more general theorem expressed in the same words.

But it is less easy to see what the corresponding theorem is, when instead of being a point-pair, the Absolute is a proper conic; and the discussion of the question affords some interesting results.

Take $(x=0,\ y=0,\ z=0)$ for the equations of the sides of the triangle, and let the equation of the Absolute be

$$(a,\ b,\ c,\ f,\ g,\ h \between x,\ y,\ z)^2=0,$$

then any two lines which are harmonics in regard to this conic (or, what is the same thing, which are such that the one of them passes through the pole of the other) are said to be perpendicular to each other, and the question is:

Find the locus of a point, such that the perpendiculars let fall therefrom on the sides of the triangle have their feet in a line.

Supposing, as usual, that the inverse coefficients are $(A,\ B,\ C,\ F,\ G,\ H)$, and that K is the discriminant, the coordinates of the poles of the three sides respectively are

(A, H, G), (H, B, F), (G, F, C). Hence considering a point P, the coordinates of which are (x, y, z), and taking (X, Y, Z) for current coordinates, the equation of the perpendicular from P on the side $X = 0$ is

$$\begin{vmatrix} X, & Y, & Z \\ x, & y, & z \\ A, & H, & G \end{vmatrix} = 0,$$

and writing in this equation $X = 0$, we find

$$(\ 0\ ,\ Ay - Hx,\ Az - Gx)$$

for the coordinates of the foot of the perpendicular. For the other perpendiculars respectively, the coordinates are

$$(Bx - Hy,\ 0\ ,\ Bz - Fy),$$

and

$$(Cx - Gz,\ Cy - Fz,\ 0\ ),$$

and hence the condition in order that the three feet may lie in a line is

$$\begin{vmatrix} 0\ , & Ay - Hx, & Az - Gx \\ Bx - Hy, & 0\ , & Bz - Fy \\ Cx - Gz, & Cy - Fz, & 0 \end{vmatrix} = 0;$$

or, what is the same thing,

$$(Ay - Hx)(Bz - Fy)(Cx - Gz) + (Az - Gx)(Bx - Hy)(Cy - Fz) = 0,$$

that is

$$\begin{aligned} &2(ABC - FGH)\,xyz \\ &+ A(FH - BG)\,yz^2 + A(FG - CH)\,y^2z \\ &+ B(FG - CH)\,zx^2 + B(GH - AF)\,z^2x \\ &+ C(GH - AF)\,xy^2 + C(HF - BG)\,x^2y = 0, \end{aligned}$$

which is the equation of the locus of P; the locus is therefore a cubic. Writing for a moment

$$(BC - F^2,\ CA - G^2,\ AB - H^2,\ GH - AF,\ HF - BG,\ FG - CH) = (A', B', C', F', G', H'),$$

and K' for the discriminant $ABC - AF^2 -$ &c., the equation is

$$2(ABC - FGH)\,xyz + Ayz(H'y + G'z) + Bzx(H'x + F'z) + Cxy(G'x + F'y) = 0,$$

or as this may also be written

$$\frac{2}{F'G'H'}(ABC - FGH)\,xyz + \frac{A}{F'}yz\left(\frac{y}{G'} + \frac{z}{H'}\right) + \frac{B}{G'}zx\left(\frac{x}{F'} + \frac{z}{H'}\right) + \frac{C}{H'}xy\left(\frac{x}{F'} + \frac{y}{G'}\right) = 0,$$

that is

$$\left[\frac{2}{F'G'H'}(ABC-FGH)-\frac{A}{F'^2}-\frac{B}{G'^2}-\frac{C}{H'^2}\right]xyz+\left(\frac{A}{F'}yz+\frac{B}{G'}zx+\frac{C}{H'}xy\right)\left(\frac{x}{F'}+\frac{y}{G'}+\frac{z}{H'}\right)=0,$$

and the cubic will therefore break up into a line and conic if only

$$\frac{2}{F'G'H'}(ABC-FGH)-\frac{A}{F'^2}-\frac{B}{G'^2}-\frac{C}{H'^2}=0,$$

and it is easy to see that conversely this is the necessary and sufficient condition in order that the cubic may so break up.

The condition is

$$\Omega=2F'G'H'(ABC-FGH)-AG'^2H'^2-BH'^2F'^2-CF'^2G'^2=0,$$

we have

$$AA'+BB'+CC'=3ABC-AF^2-BG^2-CH^2, =K'+2(ABC-FGH),$$

and thence

$$\Omega=F'G'H'(AA'+BB'+CC'-K')-AG'^2H'^2-BH'^2F'^2-CF'^2G'^2,$$

that is

$$\begin{aligned}\Omega&=-AG'H'(G'H'-A'F')-BH'F'(H'F'-B'G')-CF'G'(F'G'-C'H')-K'F'G'H',\\&=-AG'H'K'F-BH'F'K'G-CF'G'K'H-K'F'G'H',\\&=-K'(AFG'H'+BGH'F'+CHF'G'+F'G'H'),\end{aligned}$$

so that the condition $\Omega=0$ is satisfied if $K'=0$, that is if the equation

$$(A, B, C, F, G, H)(\xi, \eta, \zeta)^2=0,$$

which is the line-equation of the Absolute breaks up into factors; that is, if the Absolute be a point-pair.

In the case in question we may write

$$(A, B, C, F, G, H)(\xi, \eta, \zeta)^2=2(\alpha\xi+\beta\eta+\gamma\zeta)(\alpha'\xi+\beta'\eta+\gamma'\zeta),$$

that is

$$(A, B, C, F, G, H)=(2\alpha\alpha', 2\beta\beta', 2\gamma\gamma', \beta\gamma'+\beta'\gamma, \gamma\alpha'+\gamma'\alpha, \alpha\beta'+\alpha'\beta),$$

whence also, putting for shortness,

$$(\beta\gamma'-\beta'\gamma, \gamma\alpha'-\gamma'\alpha, \alpha\beta'-\alpha'\beta)=(\lambda, \mu, \nu),$$

we have

$$(A', B', C', F', G', H')=-(\lambda^2, \mu^2, \nu^2, \mu\nu, \nu\lambda, \lambda\mu),$$

and also

$$K'=0,\ 2(ABC-FGH)=AA'+BB'+CC', =-2(\alpha\alpha'\lambda^2+\beta\beta'\mu^2+\gamma\gamma'\nu^2).$$

The original cubic equation is

$$(\alpha\alpha'\lambda^2+\beta\beta'\mu^2+\gamma\gamma'\nu^2)\,xyz+\alpha\alpha'\lambda yz\,(\mu y+\nu z)+\beta\beta'\mu zx\,(\lambda x+\nu z)+\gamma\gamma'\nu xy\,(\lambda x+\mu y)=0,$$

and this in fact is

$$(\alpha\alpha'\lambda yz+\beta\beta'\mu zx+\gamma\gamma'\nu xy)\,(\lambda x+\mu y+\nu z)=0.$$

The equation $\lambda x+\mu y+\nu z=0$ is that of the line through the two points which constitute the Absolute; the other factor gives

$$\alpha\alpha'\lambda yz+\beta\beta'\mu zx+\gamma\gamma'\nu xy=0,$$

which is the equation of a conic through the angles of the triangle ($x=0$, $y=0$, $z=0$), and which also passes through the two points of the Absolute; in fact, writing (α, β, γ) for (x, y, z) the equation becomes $\alpha\beta\gamma\,(\alpha'\lambda+\beta'\mu+\gamma'\nu)=0$, and so also writing $(\alpha', \beta', \gamma')$ for (x, y, z) it becomes $\alpha'\beta'\gamma'\,(\alpha\lambda+\beta\mu+\gamma\nu)=0$, which relations are identically satisfied by the values of (λ, μ, ν). Hence we see that the Absolute being a point-pair, the locus is the conic passing through the angles of the triangle, and the two points of the Absolute; that is, it is the *circle* passing through the angles of the triangle.

But assuming that K' is not $=0$, or that the Absolute is a proper conic, the equation $\Omega=0$ will be satisfied if

$$AFG'H'+BGH'F'+CHF'G'+F'G'H'=0,$$

we have F', G', $H'=Kf$, Kg, Kh respectively, or omitting the factor K^2, the equation becomes

$$AFgh+BGhf+CHfg+Kfgh=0,$$

which is

$$f^2g^2h^2-bcg^2h^2-cah^2f^2\div abf^2g^2+2abcfgh=0,$$

or, as it may also be written,

$$abcf^2g^2h^2\left(\frac{1}{abc}-\frac{1}{af^2}-\frac{1}{bg^2}-\frac{1}{ch^2}+\frac{2}{fgh}\right)=0.$$

I remark that we have $ABC-FGH=K\,(abc-fgh)$; substituting also for F', G', H' the values Kf, Kg, Kh, the equation of the cubic curve is

$$2\,(abc-fgh)\,xyz+Ayz\,(hy+gz)+Bzx\,(hx+fy)+Cxy\,(gx+fy)=0,$$

and the transformed form is

$$\left[\frac{2}{fgh}(abc-fgh)-\frac{A}{f^2}-\frac{B}{g^2}-\frac{C}{h^2}\right]xyz+\left(\frac{A}{f}yz+\frac{B}{g}zx+\frac{C}{h}xy\right)\left(\frac{x}{f}+\frac{y}{g}+\frac{z}{h}\right)=0\,;$$

we have

$$\frac{2}{fgh}(abc-fgh)-\frac{A}{f^2}-\frac{B}{g^2}-\frac{C}{h^2}=abc\left(\frac{1}{abc}-\frac{1}{af^2}-\frac{1}{bg^2}-\frac{1}{ch^2}+\frac{2}{fgh}\right),$$

so that the foregoing condition

$$\frac{1}{abc}-\frac{1}{af^2}-\frac{1}{bg^2}-\frac{1}{ch^2}+\frac{2}{fgh}=0,$$

being satisfied, the cubic breaks up into the line $\frac{x}{f}+\frac{y}{g}+\frac{z}{h}=0$, and the conic

$$\frac{A}{f}yz+\frac{B}{g}zx+\frac{C}{h}xy=0.$$

It is to be remarked that in general a triangle and the reciprocal triangle are in perspective; that is, the lines joining corresponding angles meet in a point, and the points of intersections of opposite sides lie in a line; this is the case therefore with the triangle $(x=0,\ y=0,\ z=0)$, and the reciprocal triangle

$$(ax+hy+gz=0,\ hx+by+fz=0,\ gx+fy+cz=0);$$

and it is easy to see that the line through the points of intersection of corresponding sides is in fact the above mentioned line $\frac{x}{f}+\frac{y}{g}+\frac{z}{h}=0$. It is to be noticed also that the coordinates of the point of intersection of the lines joining the corresponding angles are $(F,\ G,\ H)$. The conic

$$\frac{A}{f}yz+\frac{B}{g}zx+\frac{C}{h}xy=0$$

is of course a conic passing through the angles of the triangle $(x=0,\ y=0,\ z=0)$; it is *not*, what it might have been expected to be, a conic having double contact with the Absolute $(a,\ b,\ c,\ f,\ g,\ h\between x,\ y,\ z)^2$.

I return to the condition

$$\frac{1}{abc}-\frac{1}{af^2}-\frac{1}{bg^2}-\frac{1}{ch^2}+\frac{2}{fgh}=0,$$

this can be shown to be the condition in order that the sides of the triangle $(x=0, y=0, z=0)$, and the sides of the reciprocal triangle $(ax+hy+gz=0,\ hx+by+fz=0,\ gx+fy+cz=0)$ touch one and the same conic; in fact, using line coordinates, the coordinates of the first three sides are $(1,\ 0,\ 0)$, $(0,\ 1,\ 0)$, $(0,\ 0,\ 1)$ respectively, and those of the second three sides are $(a,\ h,\ g)$, $(h,\ b,\ f)$, $(g,\ f,\ c)$ respectively; the equation of a conic touching the first three lines is

$$\frac{L}{\xi}+\frac{M}{\eta}+\frac{N}{\zeta}=0,$$

and hence making the conic touch the second three sides, we have three linear equations from which eliminating $L,\ M,\ N$, we find

$$\begin{vmatrix} \frac{1}{a}, & \frac{1}{h}, & \frac{1}{g} \\ \frac{1}{h}, & \frac{1}{b}, & \frac{1}{f} \\ \frac{1}{g}, & \frac{1}{f}, & \frac{1}{c} \end{vmatrix}=0,$$

which is the equation in question.

We know that if the sides of two triangles touch one and the same conic, their angles must lie in and on the same conic. The coordinates of the angles are (1, 0, 0), (0, 1, 0), (0, 0, 1) and (A, H, G), (H, B, F), (G, F, C) respectively, and the angles will be situate in a conic if only

$$\begin{vmatrix} \frac{1}{A}, & \frac{1}{H}, & \frac{1}{G} \\ \frac{1}{H}, & \frac{1}{B}, & \frac{1}{F} \\ \frac{1}{G}, & \frac{1}{F}, & \frac{1}{C} \end{vmatrix} = 0,$$

an equation which must be equivalent to the last preceding one; this is easily verified. In fact, writing for shortness

$$\nabla = \begin{vmatrix} \frac{1}{a}, & \frac{1}{h}, & \frac{1}{g} \\ \frac{1}{h}, & \frac{1}{b}, & \frac{1}{f} \\ \frac{1}{g}, & \frac{1}{f}, & \frac{1}{c} \end{vmatrix}, \quad \square = \begin{vmatrix} \frac{1}{A}, & \frac{1}{H}, & \frac{1}{G} \\ \frac{1}{H}, & \frac{1}{B}, & \frac{1}{F} \\ \frac{1}{G}, & \frac{1}{F}, & \frac{1}{C} \end{vmatrix},$$

we have

$$-\square = \frac{1}{ABCF^2}(BC - F^2) + \frac{1}{CFGH^2}(FG - CH) + \frac{1}{BFG^2H}(HF - BG),$$

$$= \frac{K}{ABCF^2G^2H^2}(aG^2H^2 + hABFG + gCAHF),$$

and the second factor is

$$= aGH(AF + Kf) + AFhBG + AFgCH,$$

$$= AF(aGH + hBG + gCH) + KafGH.$$

But

$$\begin{aligned} aGH + hBG + gCH &= G(aH + hB) + gCH = G - gF + gCH, \\ &= G - gF + gCH, \\ &= -g(FG - CH), \\ &= -ghK, \end{aligned}$$

so that the second factor is

$$= K(afGH - ghAF),$$

which is

$$= K(f^2g^2h^2 - bcg^2h^2 - cah^2f^2 - abf^2g^2 + 2abcfgh),$$

$$= Kabcf^2g^2h^2\left(\frac{1}{abc} - \frac{1}{af^2} - \frac{1}{bg^2} - \frac{1}{ch^2} + \frac{2}{fgh}\right),$$

$$= Kabcf^2g^2h^2\nabla,$$

so that we have identically

$$-ABCF^2G^2H^2\,\square = K^2abcf^2g^2h^2\nabla,$$

and the conditions $\nabla=0$, $\square=0$ are consequently equivalent.

The condition

$$\frac{1}{abc}-\frac{1}{af^2}-\frac{1}{bg^2}-\frac{1}{ch^2}+\frac{2}{fgh}=0,$$

is the condition in order that the function

$$\left(\frac{1}{a},\ \frac{1}{b},\ \frac{1}{c},\ \frac{1}{f},\ \frac{1}{g},\ \frac{1}{h}\between ax,\ by,\ cz\right)^2,$$

may break up into linear factors; the function in question is

$$\left(a,\ b,\ c,\ \frac{bc}{f},\ \frac{ca}{g},\ \frac{ab}{h}\between x,\ y,\ z\right)^2,$$

which is

$$=(a,\ b,\ c,\ f,\ g,\ h\between x,\ y,\ z)^2+2\left(\frac{A}{f}yz+\frac{B}{g}zx+\frac{C}{h}xy\right),$$

so that the condition is, that the conic

$$(a,\ b,\ c,\ f,\ g,\ h\between x,\ y,\ z)^2+2\left(\frac{A}{f}yz+\frac{B}{g}zx+\frac{C}{h}xy\right)=0,$$

(which is a certain conic passing through the intersections of the Absolute $(a,\ b,\ c,\ f,\ g,\ h\between x,\ y,\ z)^2=0$, and of the locus conic $\frac{A}{f}yz+\frac{B}{g}zx+\frac{C}{h}xy=0$) shall be a pair of lines. Writing the equation of the conic in question under the form

$$\left(a,\ b,\ c,\ \frac{bc}{f},\ \frac{ca}{g},\ \frac{ab}{h}\between x,\ y,\ z\right)^2=0,$$

the inverse coefficients A', B', C', F', G', H' of this conic, are

$$\left(-\frac{Abc}{f^2},\ -\frac{Bca}{g^2},\ -\frac{Cab}{h^2},\ -\frac{abc}{fgh}F,\ -\frac{abc}{fgh}G,\ -\frac{abc}{fgh}H\right),$$

so that we have $F':G':H'=F:G:H$. Hence, if in regard to this new conic we form the reciprocal of the triangle ($x=0$, $y=0$, $z=0$), and join the corresponding angles of the two triangles, the joining lines meet in a point which is the same point as is obtained by the like process from the triangle and its reciprocal in regard to the Absolute. But I do not further pursue this part of the theory.

It is to be noticed that the conic

$$\frac{A}{f}yz+\frac{B}{g}zx+\frac{C}{h}xy=0,$$

contains the angles of the reciprocal triangle, and is thus in fact the conic in which are situate the angles of the two triangles. For the coordinates of one of the angles of the reciprocal triangle are (A, H, G); we should therefore have

$$\frac{A}{f}HG + \frac{B}{g}GA + \frac{C}{h}AH = 0,$$

which is

$$\frac{A}{fgh}(GHgh + BGhf + CHfg) = 0,$$

or attending only to the second factor and writing

$$GH = Kf + AF,$$

the condition is

$$Kfgh + AFgh + BGhf + CHfg = 0,$$

or substituting for K, A, B, C, F, G, H their values and reducing, this is

$$-abcf^2g^2h^2\left(\frac{1}{abc} - \frac{1}{af^2} - \frac{1}{bg^2} - \frac{1}{ch^2} + \frac{2}{fgh}\right) = 0,$$

which is satisfied: hence the three angles of the reciprocal triangle lie on the conic in question.

Partially recapitulating the foregoing results, we see in the case where the Absolute is not a point-pair, that the locus of a point such that the perpendiculars from it on the sides of the triangle have their feet in a line, is in general a *cubic curve* passing through the angles of the triangle: if, however, the condition

$$\frac{1}{abc} - \frac{1}{af^2} - \frac{1}{bg^2} - \frac{1}{ch^2} + \frac{2}{fgh} = 0$$

be satisfied, that is, if the triangle be such that the angles thereof and of the reciprocal triangle lie in a conic (or, what is the same thing, if the sides touch a conic) then the cubic locus breaks up into the line $\frac{x}{f} + \frac{y}{g} + \frac{z}{h} = 0$, which is the line through the points of intersection of the corresponding sides of the two triangles, and into the conic

$$\frac{A}{f}yz + \frac{B}{g}zx + \frac{C}{h}xy = 0,$$

which is the conic through the angles of the two triangles.

The question arises, given a conic (the Absolute) to construct a triangle such that its angles, and the angles of the reciprocal triangle in regard to the given conic, lie in a conic.

I suppose that two of the angles of the triangle are given, and I enquire into the locus of the remaining angle. To fix the ideas, let A, B, C be the angles of the triangle, A', B', C' those of the reciprocal triangle; and let the angles A and B be given. We have to find the locus of the point C: I observe however, that the lines AA', BB', CC' meet in a point O, and I conduct the investigation in such manner as to obtain simultaneously the loci of the two points C and O. The lines $C'B'$, $C'A'$ are the polars of A, B respectively, let their equations be $x=0$, and $y=0$, and let the equation of the line AB be $z=0$; this being so, the equation of the given conic will be of the form

$$(a,\ b,\ c,\ 0,\ 0,\ h \between x,\ y,\ z)^2=0.$$

I take (α, β, γ) for the coordinates of O and (x, y, z) for those of C; the coordinates of either of these points being of course deducible from those of the other.

Observing that the inverse coefficients are

$$(bc,\ ca,\ ab-h^2,\ 0,\ 0,\ -ch),$$

we find

$$\begin{array}{lll} \text{coordinates of } A \text{ are} & (\ \ b, & -h,\ 0), \\ \quad\text{,,} \qquad\quad B\ \ \text{,,} & (-h, & \ \ a,\ 0). \end{array}$$

The points A' and B' are then given as the intersections of AO with $C'A'(y=0)$ and of BO with $C'B'(x=0)$; we find

$$\begin{array}{llll} \text{coordinates of } A' \text{ are} & (h\alpha+b\beta, & 0\quad, & h\gamma), \\ \quad\text{,,} \qquad\quad B'\ \ \text{,,} & (\quad 0\quad, & a\alpha+h\beta, & h\gamma). \end{array}$$

Moreover,

$$\begin{array}{ll} \text{coordinates of } C' \text{ are} & (0,\ 0,\ 1), \\ \quad\text{,,} \qquad\quad C\ \ \text{,,} & (x,\ y,\ z). \end{array}$$

The six points A, B, C, A', B', C' are to lie in a conic; the equations of the lines $C'A$, $C'B$, AB are $hX+bY=0$, $aX+hY=0$, $Z=0$, and hence the equation of a conic passing through the points C', A, B is

$$\frac{L}{aX+hY}+\frac{M}{hX+bY}+\frac{N}{Z}=0.$$

Hence, making the conic pass through the remaining points A', B', C, we find

$$\frac{L}{a(h\alpha+b\beta)}+\frac{M}{h(h\alpha+b\beta)}+\frac{N}{h\gamma}=0,$$

$$\frac{L}{h(a\alpha+h\beta)}+\frac{M}{b(a\alpha+h\beta)}+\frac{N}{h\gamma}=0,$$

$$\frac{L}{ax+hy}+\frac{M}{hx+by}+\frac{N}{z}=0,$$

and eliminating the L, M, N, we find

$$\begin{vmatrix} \dfrac{1}{a}\ , & \dfrac{1}{h}\ , & h\alpha + b\beta \\ \dfrac{1}{h}\ , & \dfrac{1}{b}\ , & a\alpha + h\beta \\ \dfrac{1}{ax+hy}\ , & \dfrac{1}{hx+by}\ , & \dfrac{hy}{z} \end{vmatrix} = 0,$$

or developing and reducing, this is

$$-\frac{(ab-h^2)}{hab}\frac{y}{z} + \frac{1}{h}\frac{a\alpha + h\beta}{ax+hy} + \frac{1}{h}\frac{h\alpha + b\beta}{hx+by}$$
$$-\frac{1}{a}\frac{a\alpha + h\beta}{hx+by} - \frac{1}{b}\frac{h\alpha + b\beta}{ax+hy} = 0.$$

We have still to find the relation between (α, β, γ) and (x, y, z); this is obtained by the consideration that the line $A'B'$, through the two points A', B' the coordinates of which are known in terms of (α, β, γ), is the polar of the point C, the coordinates of which are (x, y, z). The equation of $A'B'$ is thus obtained in the two forms

$$(a\alpha + h\beta)\,X + (h\alpha + b\beta)\,Y - \frac{(a\alpha + h\beta)(h\alpha + b\beta)}{hy}\,Z = 0,$$

and

$$(ax + hy)\,X + (hx + by)\,Z + \qquad cz \qquad Z = 0,$$

and comparing these, we have

$$x : y : z = \alpha : \beta : \frac{-(a\alpha + h\beta)(h\alpha + b\beta)}{chy},$$

or what is the same thing

$$\alpha : \beta : \gamma = x : y : \frac{-(ax+hy)(hx+by)}{chz},$$

(where it is to be observed that the equation $\alpha : \beta = x : y$ is the verification of the theorem that the lines AA', BB', CC' meet in a point O).

We may now from the above found relation eliminate either the (α, β, γ) or the (x, y, z); first eliminating the (α, β, γ), we find

$$-\frac{ab-h^2}{hab}\frac{Y}{Z} + \frac{2}{h} - \frac{1}{a}\frac{ax+hy}{hx+by} - \frac{1}{b}\frac{hx+by}{ax+by} = 0,$$

where

$$\frac{Y}{Z} = -\frac{(ax+hy)(hx+by)}{chz^2},$$

or, completing the elimination,

$$\frac{ab-h^2}{ch}\,\frac{(ax+hy)^2(hx+by)^2}{z^2} = (hb,\ -ab,\ ha \between ax+hy,\ hx+by)^2 = 0,$$

which is a quartic curve having a node at each of the points

$$(z=0,\ ax+hy=0),\ (z=0,\ hx+by=0),\ (ax+hy=0,\ hx+by=0),$$

that is, at each of the points B, A, C'. The right-hand side of the foregoing equation is

$$= -(ab-h^2)(ha,\ ab,\ hb\between x,\ y)^2,\ = -(ab-h^2)\,h\left(ax^2+by^2+\frac{2ab}{h}xy\right),$$

so that the equation may also be written

$$(ax+hy)^2(hx+by)^2+ch^2z^2\left(ax^2+by^2+\frac{2ab}{h}xy\right)=0.$$

Secondly, to eliminate the $(x,\ y,\ z)$, we have

$$-\frac{ab-h^2}{hab}\frac{Y}{Z}+\frac{2}{h}-\frac{1}{a}\frac{a\alpha+h\beta}{h\alpha+b\beta}-\frac{1}{b}\frac{h\alpha+b\beta}{a\alpha+h\beta}=0,$$

where

$$\frac{Y}{Z}=-\frac{chy^2}{(a\alpha+h\beta)(h\alpha+b\beta)},$$

or, completing the elimination,

$$\begin{aligned}(ab-h^2)\,chy^2 &= \quad (hb,\ -ab,\ ha\between a\alpha+h\beta,\ h\alpha+b\beta)^2\\ &= -(ab-h^2)\,h\left(a\alpha^2+b\beta^2+\frac{2ab}{h}\alpha\beta\right),\end{aligned}$$

that is

$$\left(a,\ b,\ c,\ 0,\ 0,\ \frac{ab}{h}\between \alpha,\ \beta,\ \gamma\right)^2=0.$$

Writing $(x,\ y,\ z)$ in place of $(\alpha,\ \beta,\ \gamma)$, the locus of the point O is the conic

$$\left(a,\ b,\ c,\ 0,\ 0,\ \frac{ab}{h}\between x,\ y,\ z\right)^2=0,$$

which is a conic intersecting the Absolute

$$(a,\ b,\ c,\ 0,\ 0,\ h\between x,\ y,\ z)^2=0,$$

at its intersections with the lines $x=0$, $y=0$, that is the lines $C'B'$ and $C'A'$.

In regard to this new conic, the coordinates of the pole of $C'B'$ $(x=0)$ are at once found to be $(-h,\ a,\ 0)$, that is, the pole of $C'B'$ is B; and similarly the coordinates of the pole of $C'A'$ $(y=0)$ are $(b,\ -h,\ 0)$, that is, the pole of $C'A'$ is A. We may consequently construct the conic the locus of O, viz. given the Absolute and the points A and B, we have $C'A'$ the polar of B, meeting the Absolute in two points $(a_1,\ a_2)$, and $C'B'$ the polar of A meeting the Absolute in the points (b_1 and b_2); the lines $C'A'$ and $C'B'$ meet in C'. This being so, the required conic passes through the points a_1, a_2, b_1, b_2, the tangents at these points being Aa_1, Aa_2, Bb_1, Bb_2 respectively; eight conditions, five of which would be sufficient to determine the conic. It is to be remarked that the lines $C'B'$, $C'A'$ (which in regard to the Absolute are the polars of A, B respectively) are in regard to the required conic the polars of B, A respectively.

The conic the locus of O being known, the point O may be taken at any point of this conic, and then we have A' as the intersection of $C'A'$ with AO, B' as the intersection of $C'B'$ with BO, and finally, C as the pole of the line $A'B'$ in regard to the Absolute, the point so obtained being a point on the line $C'O$. To each position of O on the conic locus, there corresponds of course a position of C; the locus of C is, as has been shown, a quartic curve having a node at each of the points C', A, B.

The foregoing conclusions apply of course to spherical figures; we see therefore that on the sphere the locus of a point such that the perpendiculars let fall on the sides of a given spherical triangle have their feet in a line (great circle), is a spherical cubic. If, however, the spherical triangle is such that the angles thereof and the poles of the sides (or, what is the same thing, the angles of the polar triangle) lie on a spherical conic; then the cubic locus breaks up into a line (great circle), which is in fact the circle having for its pole the point of intersection of the perpendiculars from the angles of the triangle on the opposite sides respectively, and into the before-mentioned spherical conic. Assuming that the angles A and B are given, the above-mentioned construction, by means of the point O, is applicable to the determination of the locus of the remaining angle C, in order that the spherical triangle ABC may be such that the angles and the poles of the sides lie on the same spherical conic, but this requires some further developments. The lines $C'B'$, $C'A'$ which are the polars of the given angles A, B respectively, are the cyclic arcs of the conic the locus of O, or say for shortness the conic O; and moreover these same lines $C'B'$, $C'A'$ are in regard to the conic O, the polars of the angles B, A respectively. If instead of the conic O we consider the polar conic O', it follows that A, B are the foci, and $C'A'$, $C'B'$ the corresponding directrices of the conic O'. The distance of the directrix $C'A'$ from the centre of the conic, measuring such distance along the transverse axis is clearly $=90° -$ distance of the focus A; it follows that the transverse semi-axis is $=45°$, or what is the same thing, that the transverse axis is $=90°$; that is, the conic O' is a conic described about the foci A, B with a transverse axis (or sum or difference of the focal distances) $=90°$. Considering any tangent whatever of this conic, the pole of the tangent is a position of the point O, which is the point of intersection of the perpendiculars let fall from the angles of the spherical triangle on the opposite sides; hence, to complete the construction, we have only through A and B respectively to draw lines AC, BC perpendicular to the lines BO, CO respectively; the lines in question will meet in a point C, which is such that CO will be perpendicular to AB, and which point C is the required third angle of the spherical triangle ABC. In order to ascertain whether a given spherical triangle ABC has the property in question (viz. whether it is such that the angles thereof and of the polar triangle lie in a spherical conic), we have only to construct as before the conic O' with the foci A, B and transverse axis $=90°$, and then ascertain whether the polar of the point O, the intersections of the perpendiculars from the angles of the triangle on the opposite sides respectively, is a tangent of the conic O'. It is moreover clear, that given a triangle ABC having the property in question, if with the foci A, B and transverse axis $=90°$ we describe a conic, and if in like manner with the foci A, C and the same transverse axis, and with the foci B, C and the same transverse axis, we describe two other conics; then that the three conics will have a common tangent the pole whereof will be the point of intersection of the perpendiculars from the angles of the triangle ABC on the opposite sides respectively.

395.

INVESTIGATIONS IN CONNEXION WITH CASEY'S EQUATION.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. VIII. (1867), pp. 334—341.]

In a paper read April 9, 1866, and recently published in the *Proceedings of the Royal Irish Academy*, Mr Casey has given in a very elegant form the equation of a pair of circles touching each of three given circles, viz. if $U=0$, $V=0$, $W=0$ be the equations of the three given circles respectively, and if considering the common tangents of ($V=0$, $W=0$), of ($W=0$, $U=0$), and of ($U=0$, $V=0$) respectively, these common tangents being such that the centres of similitude through which they respectively pass lie in a line (viz. the tangents are all three direct, or one is direct and the other two are inverse), then if f, g, h are the lengths of the tangents in question, the equation

$$\sqrt{(fU)}+\sqrt{(gV)}+\sqrt{(hW)}=0,$$

belongs to a pair of circles, each of them touching the three given circles. (There are, it is clear, four combinations of tangents, and the theorem gives therefore the equations of four pairs of circles, that is of the eight circles which touch the three given circles.)

Generally, if $U=0$, $V=0$, $W=0$ are the equations of any three curves of the same order n, and if f, g, h are arbitrary coefficients, then the equation

$$\sqrt{(fU)}+\sqrt{(gV)}+\sqrt{(hW)}=0,$$

is that of a curve of the order $2n$, touching each of the curves $U=0$, $V=0$, $W=0$, n^2 times, viz. it touches

$$\begin{array}{llll} U=0, & \text{at its } n^2 \text{ intersections with} & & gV-hW=0, \\ V=0 & ,, & ,, & hW-fU=0, \\ W=0 & ,, & ,, & fU-gV=0. \end{array}$$

If however the curves $U=0$, $V=0$, $W=0$ have a common intersection, then the curve in question has a node at this point, and besides touches each of the three curves in n^2-1 points; and similarly, if the curves $U=0$, $V=0$, $W=0$ have k common intersections, then the curve in question has a node at each of these points, and besides touches each of the three curves in n^2-k points.

In particular, if $U=0$, $V=0$, $W=0$ are conics having two common intersections, then the curve is a quartic having a node at each of the common intersections, and besides touching each of the given conics in two points; whence, if the coefficients f, g, h (that is, their ratios) are so determined that the quartic may have two more nodes, then the quartic, having in all four nodes, will break up into a pair of conics, each passing through the common intersections, and the pair touching each of the given conics in two points; that is, the component conics will each of them touch each of the given conics once. Taking the circular points at infinity for the common intersections, the conics will be circles, and we thus see that Casey's theorem is in effect a determination of the coefficients f, g, h, in such wise that the curve

$$\surd(fU)+\surd(gV)+\surd(hW)=0,$$

(which when $U=0$, $V=0$, $W=0$ are circles, is by what precedes a bicircular quartic) shall have two more nodes, and so break up into a pair of circles.

The question arises, given $U=0$, $V=0$, $W=0$, curves of the same order n, it is required to determine the ratios $f:g:h$ in such wise that the curve

$$\surd(fU)+\surd(gV)+\surd(hW)=0,$$

may have two nodes; or we may simply inquire as to the number of the sets of values of $(f:g:h)$, which give a binodal curve, $\surd(fU)+\surd(gV)+\surd(hW)=0$.

I had heard of Mr Casey's theorem from Dr Salmon, and communicated it together with the foregoing considerations to Prof. Cremona, who, in a letter dated Bologna, March 3, 1866, sent me an elegant solution of the question as to the number of the binodal curves. This solution is in effect as follows:

LEMMA. Given the curves $U=0$, $V=0$, $W=0$ of the same order n; consider the point (f, g, h), and corresponding thereto the curve $fU+gV+hW=0$. As long as the point (f, g, h) is arbitrary, the curve $fU+gV+hW=0$, will not have any node, and in order that this curve may have a node, it is necessary that the point (f, g, h) shall lie on a certain curve Σ; this being so, the node will lie on a curve J, the Jacobian of the curves U, V, W; and the curves J and Σ will correspond to each other, point to point; viz. taking for (f, g, h) any point whatever on the curve Σ, the curve $fU+gV+hW=0$ will be a curve having a node at some one point on the curve J; and conversely, in order that the curve $fU+gV+hW=0$ may be a curve having a node at a given point on the curve J, it is necessary that the point (f, g, h) shall be at some one point of the curve Σ. The curve Σ has however nodes and cusps; each node of Σ corresponds to two points of J, viz. the point (f, g, h) being at a node of Σ, the curve $fU+gV+hW=0$, is a binodal curve having a node at each of the corresponding points on J; and each cusp of Σ corresponds to two coincident points of J, viz. the point (f, g, h) being at a cusp of Σ, the curve $fU+gV+hW=0$ is a cuspidal curve having a cusp at the corresponding point of J. The number of the binodal curves $fU+gV+hW=0$ is thus equal to the number of the nodes of Σ, and the number of the cuspidal curves $fU+gV+hW=0$ is equal to the number of the cusps of Σ. The curve Σ is easily shown to be a curve of the order $3(n-1)^2$ and class $3n(n-1)$; and quà curve which corresponds point to point with J, it is a curve having the same deficiency as J, that is a deficiency $=\frac{1}{2}(3n-4)(3n-5)$; we have thence the Plückerian numbers of the curve Σ, viz.:

Order is	$= 3(n-1)^2$,
Class	$= 3n(n-1)$,
Cusps	$= 12(n-1)(n-2)$,
Nodes	$= \frac{3}{2}(n-1)(n-2)(3n^2-3n-11)$,
Inflexions	$= 3(n-1)(4n-5)$,
Double tangents	$= \frac{3}{2}(n-1)(n-2)(3n^2+3n-8)$.

Remarks. The consideration of the foregoing curve Σ is, I believe, first due to Prof. Cremona, it is a curve related to the three distinct curves $U=0$, $V=0$, $W=0$, in the same way precisely as Steiner's curve P_0 is related to the three curves $d_xU=0$, $d_yU=0$, $d_zU=0$. (Steiner, "Allgemeine Eigenschaften der algebraischen Curven," *Crelle*, t. XLVII. (1854), pp. 1—6; see also Clebsch, "Ueber einige von Steiner behandelte Curven," *Crelle*, t. LXIV. (1865), pp. 288—293), and the Plückerian numbers of P_0 (writing therein $n+1$ for n) are identical with those of Σ. The foregoing expressions $\frac{3}{2}(n-1)(n-2)(3n^2-3n-11)$ and $12(n-1)(n-2)$ for the numbers of the binodal and cuspidal curves $fU+gV+hW=0$, are given in my memoir "On the Theory of Involution," *Cambridge Philosophical Transactions*, t. XI. (1866), pp. 21—38, see p. 32, [348]; but the employment of the curve Σ very much simplifies the investigation.

Passing now to the proposed question, we have as before the curves $U=0$, $V=0$, $W=0$, of the same order n; and we may consider the point (f, g, h), and corresponding thereto the curve $\sqrt{(fU)}+\sqrt{(gV)}+\sqrt{(hW)}=0$, say for shortness the curve Ω, which is a curve of the order $2n$, having n^2 contacts with each of the given curves U, V, W. As long as the point (f, g, h) is arbitrary, the curve Ω has not any node; and in order that this curve may have a node, it is necessary that the point (f, g, h) shall lie on a certain curve Δ; this being so, the node will lie on the foregoing curve J, the Jacobian of the given curves U, V, W; and the curves J and Δ will correspond to each other, point to point, viz. taking for (f, g, h) any point whatever on the curve Δ, the curve Ω will have a node at some one point of J; and conversely, in order that the curve Ω may be a curve having a node at a given point of J, it is necessary that the point (f, g, h) shall be at some one point of the curve Δ. The curve Δ has however nodes and cusps; each node of Δ corresponds to two points of J, viz. for (f, g, h) at a node of Δ, the curve Ω is a binodal curve having a node at each of the corresponding points of J; each cusp of Δ corresponds to two coincident points of J, viz. for (f, g, h) at a cusp of Δ, the curve Ω is a cuspidal curve having a cusp at the corresponding point of J. The number of the binodal curves Ω is consequently equal to that of the nodes of Δ, and the number of the cuspidal curves Ω is equal to that of the cusps of Δ; we have consequently to find the Plückerian numbers of the curve Δ; and this Prof. Cremona accomplishes by bringing it into connexion with the foregoing curve Σ, and making the determination depend upon that of the number of the conics which satisfy certain conditions of contact in regard to the curve Σ.

Consider, as corresponding to any given point (f, g, h) whatever, the conic $\frac{f}{x}+\frac{g}{y}+\frac{h}{z}=0$ which passes through three fixed points, the angles of the triangle $x=0$, $y=0$, $z=0$. For points (f, g, h) which lie in an arbitrary line $Af+Bg+Ch=0$, the corresponding conics pass through the fourth fixed point $x:y:z=A:B:C$. Assume for the moment that to the points (f, g, h) which lie on the foregoing curve Δ, correspond conics which touch the foregoing curve Σ. Then 1°. to the points of intersection of the curve Δ with an arbitrary line, correspond the conics which pass through four arbitrary points and touch the curve Σ; or the order of the curve Δ is equal to the number of the conics which can be drawn through four arbitrary points to touch the curve Σ; viz. if m be the order, n the class of Σ, the number of these conics is $=2m+n$, or substituting for m, n the values $3(n-1)^2$ and $3n(n-1)$ respectively, the number of these conics, that is the order of Δ, is $=3(n-1)(3n-2)$. 2°. To the nodes of Δ correspond the conics which pass through three arbitrary points and have two contacts with Σ, viz. if m be the order, n the class, and κ the number of cusps of Σ, then the number of these conics is $=\frac{1}{2}(2m+n)^2-2m-5n-\frac{3}{2}\kappa$, or substituting for m, n their values as above, and for κ its value $=12(n-1)(n-2)$, the number of these conics, that is, the number of the nodes of Δ, is found to be

$$=\tfrac{3}{2}(n-1)(27n^3-63n^2+22n+16).$$

3°. To the cusps of Δ correspond the conics which pass through three arbitrary points, and have with Σ a contact of the second order; the number of these (m, n, κ as above) is $=3n+\kappa$, or substituting for n and κ their values as above, the number of these conics, that is the number of the cusps of Δ, is $=3(n-1)(7n-8)$. We have thence all the Plückerian numbers of the curve Δ, viz. these are

Order	$= 3(n-1)(3n-2)$,
Class	$= 6(n-1)^2$,
Nodes	$= \frac{3}{2}(n-1)(27n^3-63n^2+22n+16)$,
Cusps	$= 3(n-1)(7n-8)$,
Double tangents	$= \frac{3}{2}(n-1)(12n^3-36n^2+19n+16)$,
Inflexions	$=12(n-1)(n-2)$,

and as a verification it is to be observed, that the deficiency of the curve Δ is equal to that of the curve J, viz. it has the value $\frac{1}{2}(3n-4)(3n-5)$. The foregoing numbers include the result that the number of the binodal curves

$$\sqrt{(fU)}+\sqrt{(gV)}+\sqrt{(hW)}=0,$$

is

$$=\tfrac{3}{2}(n-1)(27n^3-63n^2+22n+16).$$

The proof depended on the assumption, that to the points (f, g, h) which lie on the curve Δ, correspond the conics $\frac{f}{x}+\frac{g}{y}+\frac{h}{z}=0$ which touch the curve Σ; this M. Cremona proves in a very simple manner: the points of J correspond each to each with the points of Σ, or if we please they correspond each to each with the tangents of Σ. To the $6n(n-1)$ intersections of J with any curve Ω $\left(\text{viz. } \sqrt{(fU)}+\sqrt{(gV)}+\sqrt{(hW)}=0\right)$ correspond the $6n(n-1)$ common tangents of Σ and the conic $\frac{f}{x}+\frac{g}{y}+\frac{h}{z}=0$; if Ω has a node, two of the $6n(n-1)$ intersections coincide, and the corresponding two tangents will also coincide, that is Ω having a node (or the point (f, g, h) being on the curve Δ), the conic touches the curve Σ. But it is not uninteresting to give an independent analytical proof. Write for shortness

$$\begin{aligned} dU &= A dx + B dy + C dz, \\ dV &= A' dx + B' dy + C' dz, \\ dW &= A'' dx + B'' dy + C'' dz, \end{aligned}$$

and let (x, y, z) be the coordinates of a point on J, (X, Y, Z) those of the corresponding point on Σ, (f, g, h) those of the corresponding point on Δ. Write also for shortness

$$BC' - B'C,\ CA' - C'A,\ AB' - A'B = P : Q : R,$$

then we have

$$\begin{aligned} AX + BY + CZ &= 0, \\ A'X + B'Y + C'Z &= 0, \\ A''X + B''Y + C''Z &= 0, \end{aligned}$$

$$\begin{aligned} A\sqrt{\left(\frac{f}{U}\right)} + B\sqrt{\left(\frac{g}{V}\right)} + C\sqrt{\left(\frac{h}{W}\right)} &= 0, \\ A' \quad ,, \quad + B' \quad ,, \quad + C' \quad ,, \quad &= 0, \\ A'' \quad ,, \quad + B'' \quad ,, \quad + C'' \quad ,, \quad &= 0, \end{aligned}$$

giving $\begin{vmatrix} A, & B, & C \\ A', & B', & C' \\ A'', & B'', & C'' \end{vmatrix} = 0$, which is in fact the equation of the curve J; and moreover giving $X : Y : Z = P : Q : R$, to determine the point (X, Y, Z) on Σ; and

$$\sqrt{\left(\frac{f}{U}\right)} : \sqrt{\left(\frac{g}{V}\right)} : \sqrt{\left(\frac{h}{W}\right)} = P : Q : R,$$

or, what is the same thing, $f : g : h = P^2U : Q^2V : R^2W$, to determine the point (f, g, h) on Δ. Treating now (f, g, h) as constants, and (X, Y, Z) as current coordinates, the conic $\frac{f}{X}+\frac{g}{Y}+\frac{h}{Z}=0$, will touch the curve Σ at the point (P, Q, R), if only the equation of the conic is satisfied by these values and by the consecutive values $P+dP$, $Q+dQ$, $R+dR$; or what is the same thing, if we have

$$\frac{f}{P}+\frac{g}{Q}+\frac{h}{R}=0,$$

$$\frac{fdP}{P^2}+\frac{gdQ}{Q^2}+\frac{hdR}{R^2}=0,$$

that is

$$\frac{f}{P^2}:\frac{g}{Q^2}:\frac{h}{R^2}=QdR-RdQ : RdP-PdR : PdQ-QdP.$$

If the functions on the right-hand side are as $U : V : W$, then these equations give

$$f:g:h=P^2U : Q^2V : R^2W,$$

that is (f, g, h) will be a point on the curve Δ. It is therefore only necessary to show that in virtue of the equation $J=0$ of the curve J, and of the derived equation $dJ=0$, we have

$$QdR-RdQ : RdP-PdR : PdQ-QdP=U : V : W.$$

Take for instance the equation

$$V(QdR-RdQ)-U(RdP-PdR)=0,$$

that is

$$dR(UP+VQ+WR)-R(UdP+VdQ+WdR)=0,$$

and this, and the other two equations will be satisfied if only $UP+VQ+WR=0$, $UdP+VdQ+WdR=0$; we have, neglecting a numerical factor,

$$\begin{aligned} U &= Ax+A'y+A''z, \\ V &= Bx+B'y+B''z, \\ W &= Cx+C'y+C''z, \end{aligned}$$

whence, attending to the values of P, Q, R, we have

$$UP+VQ+WR=zJ=0;$$

hence also

$$UdP+VdQ+WdR+(PdU+QdV+RdW)=0,$$

so that

$$UdP+VdQ+WdR=0,$$

if only

$$PdU+QdV+RdW=0,$$

and substituting for P, Q, R, dU, dV, dW their values, the left-hand side is $= -Jdz$, which is $=0$; hence the equations in question are proved, and (f, g, h) is a point on the curve Δ.

It is to be noticed, that the two curves Σ, Δ are geometrically connected through the three arbitrary points as follows: viz. taking as axes the sides of the triangle formed by these three points, then starting from any point (f, g, h) of Δ, we take the inverse point $\left(\frac{1}{f}, \frac{1}{g}, \frac{1}{h}\right)$, the harmonic line thereof $fx + gy + hz = 0$, and finally the inverse conic $\frac{f}{x} + \frac{g}{y} + \frac{h}{z} = 0$, which by what precedes touches Σ in the point corresponding to the assumed point (f, g, h) of Δ: and conversely starting with an assumed point on Σ, we take the conic $\frac{f}{x} + \frac{g}{y} + \frac{h}{z} = 0$ which passes through the angles of the triangle and touches Σ at the assumed point; the inverse line $fx + gy + hz = 0$; the harmonic point $\left(\frac{1}{f}, \frac{1}{g}, \frac{1}{h}\right)$ of this line; and finally the inverse point (f, g, h), which will be on the curve Δ, the point corresponding to the assumed point on the curve Σ.

396.

ON A CERTAIN ENVELOPE DEPENDING ON A TRIANGLE INSCRIBED IN A CIRCLE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. IX. (1868), pp. 31—41 and 175—176.]

CONSIDERING a triangle and the circumscribed circle, and from any point of the circle drawing perpendiculars to the sides of the triangle; the feet of the three perpendiculars lie on a line; and (regarding the point as a variable point on the circle) the envelope of the line is a curve of the third class, having the line infinity for a double tangent, and being therefore a curve of the fourth order with three cusps, see Steiner's paper "Ueber eine besondere Curve dritter Klasse und vierten Grades," *Crelle*, t. LIII. (1857), pp. 231—237, which contains a series of very beautiful geometrical properties.

Mr Greer, in a paper in the last volume of the *Journal*, has expressed the equation of the line in a very elegant form, viz. if α, β, γ are the perpendicular distances of the point from the sides of the triangle; A, B, C the angles of the triangle; $(\lambda, \mu, \nu) = \left(\frac{\cos A}{\alpha}, \frac{\cos B}{\beta}, \frac{\cos C}{\gamma}\right)$; and (X, Y, Z) certain current coordinates, viz. these are the perpendicular distances from the sides, multiplied by $\sin A \tan A$, $\sin B \tan B$, $\sin C \tan C$ respectively; then the equation of the line is

$$X\lambda(\lambda-\mu)(\lambda-\nu) + Y\mu(\mu-\nu)(\mu-\lambda) + Z\nu(\nu-\lambda)(\nu-\mu) = 0,$$

where the parameters λ, μ, ν are connected by the equation $\lambda \tan A + \mu \tan B + \nu \tan C = 0$, or say by the equation

$$\lambda\mathrm{a} + \mu\mathrm{b} + \nu\mathrm{c} = 0.$$

We have a cubic equation in (λ, μ, ν) with coefficients which are linear functions of (X, Y, Z), and the required equation is that obtained by equating to zero the reciprocant of this cubic function, the facients of the reciprocant being the (a, b, c) of the linear relation; the reciprocant is of the degree 6 in (a, b, c) and of the degree 4 in the coefficients of the cubic function, that is in (X, Y, Z). But I remark that the equation in (λ, μ, ν), regarding these quantities as coordinates, is that of a cubic curve having a node at the point $\lambda = \mu = \nu$, or say the point (1, 1, 1); the corresponding value of $\lambda\mathrm{a} + \mu\mathrm{b} + \nu\mathrm{c}$ is $= \mathrm{a} + \mathrm{b} + \mathrm{c}$, and the reciprocant consequently contains the factor $(\mathrm{a} + \mathrm{b} + \mathrm{c})^2$, or dividing this out, the equation is only of the degree 4 in (a, b, c). The equation of the curve thus is

$$\frac{1}{(\mathrm{a}+\mathrm{b}+\mathrm{c})^2}\ \text{recip.}\ \{X\lambda(\lambda-\mu)(\lambda-\nu) + Y\mu(\mu-\nu)(\mu-\lambda) + Z\nu(\nu-\lambda)(\nu-\mu)\} = 0,$$

being of the degree 4 in (a, b, c), and also of the degree 4 in (X, Y, Z), that is, treating (X, Y, Z) as current coordinates, the envelope is as above stated a curve of the fourth order.

A symmetrical method for finding the reciprocant of a cubic function was given by Hesse, see my paper "On Homogeneous Functions of the Third Order with Three Variables," *Camb. and Dubl. Math. Jour.*, vol. I. (1846), pp. 97—104, [35]; the developed expression there given for the reciprocant is however erroneous; the correct value is given in my "Third Memoir on Quantics," *Phil. Trans.*, vol. CXLVI. (1856), see the Table 67, p. 644, [144] and we have only in the table to substitute for (ξ, η, ζ) the quantities (a, b, c), and for $(a, b, c, f, g, h, i, j, k, l)$ the coefficients of the cubic function of (λ, μ, ν), viz. multiplying by 6 in order to avoid fractions, these are

$$\begin{array}{l} (a,\ b,\ c,\ f,\ g,\ h,\ i,\ j,\ k,\ l\ ) \\ = (6X,\ 6Y,\ 6Z,\ -2Y,\ -2Z,\ -2X,\ -2Z,\ -2X,\ -2Y,\ X+Y+Z) \end{array}$$

respectively. The substitution might be performed as follows, viz. for the coefficient of a^6, we have

$$\left.\begin{array}{lllll} b^2c^2 & +1\,.\,1296 & Y^2Z^2 & + & 1296 \\ bcfi & -6\,.\ \ 144 & Y^2Z^2 & - & 864 \\ bi^3 & +4\,.-48 & YZ^3 & - & 192 \\ cf^3 & +4\,.-48 & Y^3Z & - & 192 \\ f^2i^2 & -3\,.\ \ \ 16 & Y^2Z^2 & - & 48 \end{array}\right\} = -192YZ(Y-Z)^2,$$

and so for the other coefficients; but I have not gone through the labour of performing the calculation. Omitting the numerical factor -192, the coefficients of $\mathrm{a}^6, \mathrm{b}^6, \mathrm{c}^6$ are of course

$$YZ(Y-Z)^2,\ ZX(Z-X)^2,\ XY(X-Y)^2;$$

and I find also that the coefficient of b^5c (the factor -192 being omitted) is

$$= ZX(3X^2 + 3Z^2 + 3YZ - 6ZX + 5XY),$$

whence those of the terms c^5a, &c. are also known.

I denote the result as follows:

$$(YZ(Y-Z)^2,\ ZX(Z-X)^2,\ XY(X-Y)^2,\ \ldots \between a,\ b,\ c)^6 = 0;$$

this equation divides as we have seen by $(a+b+c)^2$, and the quotient is

$$(YZ(Y-Z)^2,\ ZX(Z-X)^2,\ XY(X-Y)^2,\ \ldots \between a,\ b,\ c)^4 = 0;$$

and it may be remarked that the coefficient of b^3c in this quartic function of (a, b, c) is

$$= ZX(X^2 + Z^2 + 3YZ - 2ZX + 5XY.$$

The last mentioned equation, if the calculation were completed, would be analytically the best form for the equation of the envelope; but in view of what follows, I will change it by writing ax, by, cz in place of (X, Y, Z); x is therefore $= \dfrac{1}{\tan A} X$, that is, it is $=$ perpendicular distance $\times \sin A$; or, what is the same thing, the new coordinates (x, y, z) are proportional to the perpendicular distances from the sides, each distance divided by the perpendicular distance of the side from the opposite angle, the equation of the line infinity is thus $x + y + z = 0$. I write also $(a, b, c) = \left(\dfrac{1}{a}, \dfrac{1}{b}, \dfrac{1}{c}\right)$, that is, we have $(a, b, c) = (\cot A, \cot B, \cot C)$. The system of equations is therefore

$$\frac{x}{a}\lambda(\lambda-\mu)(\lambda-\nu) + \frac{y}{b}\mu(\mu-\nu)(\mu-\lambda) + \frac{z}{c}\nu(\nu-\lambda)(\nu-\mu) = 0,$$

$$\frac{\lambda}{a} + \frac{\mu}{b} + \frac{\nu}{c} = 0,$$

giving for the envelope the equation

$$bcyz(cy - bz)^2 + cazx(az - cx)^2 + abxy(bx - ay)^2 + \&c. = 0;$$

and in this function, corresponding to the term

$$b^3cZX(X^2 + Z^2 + 3YZ - 2ZX + 5XY),$$

we have the term

$$azx(bc^2x^2 + a^2bz^2 + 3a^2cyz - 2abczx + 5ac^2xy).$$

It may be noticed that, arranging in powers of (x, y, z), the several portions of each coefficient are distinct literal functions; thus we see that the coefficient of z^3x is $= a^3c + a^3b +$ other combinations of (a, b, c): this is material in order to the comparison of the foregoing equation of the envelope in a different form which will be presently mentioned.

I proceed to find the tangential equation of the envelope. Representing the equation of the line by

$$\xi x + \eta y + \zeta z = 0,$$

we have

$$\xi : \eta : \zeta = \frac{1}{a}\lambda(\lambda-\mu)(\lambda-\nu) : \frac{1}{b}\mu(\mu-\nu)(\mu-\lambda) : \frac{1}{c}\nu(\nu-\lambda)(\nu-\mu),$$

or, what is the same thing,

$$\xi : \eta : \zeta = \frac{1}{a}\ \frac{\lambda}{\mu-\nu} : \frac{1}{b}\ \frac{\mu}{\nu-\lambda} : \frac{1}{c}\ \frac{\nu}{\lambda-\mu},$$

where as before

$$\frac{\lambda}{a} + \frac{\mu}{b} + \frac{\nu}{c} = 0,$$

and eliminating λ, μ, ν, we find

$$a\xi(\eta-\zeta)^2 + b\eta(\zeta-\xi)^2 + c\zeta(\xi-\eta)^2 = 0.$$

In fact we find at once

$$\begin{aligned} a\xi(\eta-\zeta)^2 : b\eta(\zeta-\xi)^2 : c\zeta(\xi-\eta)^2 = {} & (\mu-\nu)\lambda\left\{\frac{1}{b}\mu(\lambda-\mu) - \frac{1}{c}\nu(\nu-\lambda)\right\} \\ & : (\nu-\lambda)\mu\left\{\frac{1}{c}\nu(\mu-\nu) - \frac{1}{a}\lambda(\lambda-\mu)\right\} \\ & : (\lambda-\mu)\nu\left\{\frac{1}{a}\lambda(\nu-\lambda) - \frac{1}{b}\mu(\mu-\nu)\right\}, \end{aligned}$$

and the sum of the three expressions on the right-hand side is

$$= -(\mu-\nu)(\nu-\lambda)(\lambda-\mu)\left(\frac{\lambda}{a} + \frac{\mu}{b} + \frac{\nu}{c}\right) = 0,$$

which verifies the result just obtained.

The tangential equation of the envelope is thus

$$a\xi(\eta-\zeta)^2 + b\eta(\zeta-\xi)^2 + c\zeta(\xi-\eta)^2 = 0,$$

or the envelope is a curve of the third class having as a double tangent the line $\xi = \eta = \zeta$, that is the line infinity; in fact for these values the equation $\xi x + \eta y + \zeta z = 0$ becomes $x + y + z = 0$, which is the equation of the line infinity. The curve is *therefore* a curve of the fourth order, the equation of which is

$$\frac{1}{(x+y+z)^2}\ \text{recip.}\ \{a\xi(\eta-\zeta)^2 + b\eta(\zeta-\xi)^2 + c\zeta(\xi-\eta)^2\} = 0,$$

where the reciprocant in question may be calculated from the before mentioned table 67, viz. multiplying by 3 in order to avoid fractions, the coefficients of the table are

$$\begin{array}{l}(a,\ b,\ c,\ f,\ g,\ h,\ i,\ j,\ k,\quad l\qquad)\\ =(0,\ 0,\ 0,\ c,\ a,\ b,\ b,\ c,\ a,\ -a-b-c)\end{array}$$

respectively, and for the facients $(\xi,\ \eta,\ \zeta)$ of the table we have to write $(x,\ y,\ z)$. The expression of the reciprocant is

$$= b^2c^2x^6 + c^2a^2y^6 + a^2b^2z^6 + \&c.,$$

and dividing by $(x+y+z)^2$ we have the equation of the envelope in the form

$$b^2c^2x^4 + c^2a^2y^4 + a^2b^2z^4 + \&c. = 0,$$

which must of course be identical with the former result

$$bcyz\,(cy-bz)^2 + cazx\,(az-cx)^2 + abxy\,(bx-ay)^2 + \&c. = 0.$$

Instead of discussing the curve of the third class

$$a\xi\,(\eta-\zeta)^2 + b\eta\,(\zeta-\xi)^2 + c\zeta\,(\xi-\eta)^2 = 0,$$

it will be convenient to write $(x,\ y,\ z)$ in place of $(\xi,\ \eta,\ \zeta)$, and discuss the curve of the third order, or cubic curve

$$U = ax\,(y-z)^2 + by\,(z-x)^2 + cz\,(x-y)^2 = 0,$$

which is of course a curve having a node at the point $(x=y=z)$, or say at the point $(1,\ 1,\ 1)$, and having therefore three inflexions lying in a line. The equation of the tangents at the node is found to be

$$a\,(y-z)^2 + b\,(z-x)^2 + c\,(x-y)^2 = 0,$$

that is, at the node the second derived functions of U are proportional to

$$(b+c,\ c+a,\ a+b,\ -a,\ -b,\ -c).$$

The equation of the Hessian may be found directly, or by means of the table, No. 61, in my memoir above referred to. It is as follows:

$$\begin{aligned}&(b+c)\,\{a\,(b+c)+2bc\}\,x^3\\ &+(c+a)\,\{b\,(c+a)+2ca\}\,y^3\\ &+(a+b)\,\{c\,(a+b)+2ab\}\,z^3\\ &-(3a^2+2bc+2ca+2ab)\,(cy^2z+byz^2)\\ &-(3b^2+2bc+2ca+2ab)\,(az^2x+czx^2)\\ &-(3c^2+2bc+2ca+2ab)\,(bx^2y+axy^2)\\ &+\{4\,(bc^2+b^2c+ca^2+c^2a+ab^2+a^2b)+6abc\}\,xyz=0.\end{aligned}$$

I find that in the function $b^2c^2x^6 + \&c.$ the term in z^5x is

$$= a(4a^2b + 4a^2c + 4ab^2 + 4abc - 2b^2c),$$

whence in the function $b^2c^2x^4 + \&c.$ the term in z^3x is

$$= z^3xa(4a^2b + 4a^2c + 2ab^2 + 4abc - 2b^2c),$$

a portion whereof is $= 4z^3x(a^3b + a^3c)$; and we thus obtain the numerical factor $= 4$, and thence the identity

$$b^2c^2x^4 + c^2a^2y^4 + a^2b^2z^4 + \&c. = 4bcyz(cy - bz)^2 + \ldots + \&c.$$

which equation I represent by

$$Ax^3 + By^3 + Cz^3 + 3(Fy^2z + Gz^2x + Hx^2y + Iyz^2 + Jzx^2 + Kxy^2) + 6Lxyz = 0,$$

or

$$(A, B, C, F, G, H, I, J, K, L \between x, y, z)^3 = 0,$$

viz. writing for shortness

$$M = bc + ca + ab,$$

the values of the coefficients are as follows:

$$\begin{array}{lll}
A = 3(b+c)(bc+M), & F = -c(3a^2+M), & I = -b(3a^2+M), \\
B = 3(c+a)(ca+M), & G = -a(3b^2+M), & J = -c(3b^2+M), \\
C = 3(a+b)(ab+M), & H = -b(3c^2+M), & K = -a(3c^2+M),
\end{array}$$

$$L = 2(a+b+c)M - 3abc.$$

I remark that the cubic having a node at the point (1, 1, 1), the Hessian has at this point a node with the same tangents. The second derived functions for the Hessian are therefore at the node proportional to those of the cubic; it is easy to verify that we have in fact

$$\begin{array}{ll}
A + H + J = (b+c)M, & L + F + I = -aM, \\
K + B + F = (c+a)M, & J + L + G = -bM, \\
G + I + C = (a+b)M, & H + K + L = -cM,
\end{array}$$

these values give also

$$\begin{array}{l}
A + K + G + 2L + 2J + 2H = 0, \\
H + B + I + 2F + 2L + 2K = 0, \\
J + F + C + 2I + 2G + 2L = 0,
\end{array}$$

equations which merely express that the first derived functions vanish at the node. If, by these equations we express A, B, C in terms of the other coefficients, and substitute these values in the equation of the Hessian, this may be expressed in the form

$$\begin{array}{l}
(y-z)^2\{Lx + (2F + I + L)y + (F + 2I + L)z\} \\
+ (z-x)^2\{(G + 2J + L)x + Ly + (2G + J + L)z\} \\
+ (x-y)^2\{(2H + K + L)x + (H + 2K + L)y + Lz\} = 0,
\end{array}$$

a form which puts in evidence the node (1, 1, 1).

I write

$$X = y - z, \quad Y = z - x, \quad Z = x - y,$$

so that we have identically

$$X + Y + Z = 0,$$

and that the equation of the tangents at the node is

$$aX^2 + bY^2 + cZ^2 = 0.$$

I write also for shortness

$$\begin{aligned} F' &= 2F + I + L, & I' &= F + 2I + L, \\ G' &= 2G + J + L, & J' &= G + 2J + L, \\ H' &= 2H + K + L, & K' &= H + 2K + L, \end{aligned}$$

the equation of the cubic is then

$$U = axX^2 + byY^2 + czZ^2, = 0,$$

and that of the Hessian is

$$\tilde{H}U = X^2(Lx + F'y + I'z) + Y^2(J'x + Ly + G'z) + Z^2(H'x + K'y + Lz).$$

Now observing that we have

$$\begin{aligned} L + 3aM &= L - 3(I + F + L) = -F' - I', \\ L + 3bM &= L - 3(G + J + L) = -G' - J', \\ L + 3cM &= L - 3(H + K + L) = -H' - K', \end{aligned}$$

we find

$$\begin{aligned} \tilde{H}U + 3MU = \quad & X^2(I'Y - F'Z) \\ & + Y^2(J'Z - G'X) \\ & + Z^2(K'X - H'Y), \end{aligned}$$

which shows that the function $\tilde{H}U + 3MU$ is a cubic function of $y - z$, $z - x$, $x - y$, decomposable therefore into three linear factors; and the equation $\tilde{H}U + 3MU = 0$, is consequently that of the three lines drawn from the node to the three inflexions of the cubic (or the Hessian). We know also that the Hessian of the three lines is the pair of tangents at the node([1]), viz. that regarding any one of the variables X, Y, Z as a linear function of the third of them (in virtue of the equation $X + Y + Z = 0$), then that the cubic function of X, Y, Z has $aX^2 + bY^2 + cZ^2$ for its Hessian.

[1] Taking as the canonical form of a nodal cubic $U = x^3 + y^3 + 6lxyz = 0$, then we have $\tilde{H}U = x^3 + y^3 - 2lxyz = 0$; $x^3 + y^3 = 0$ is the equation of the lines from the node to the inflexions, and the Hessian of the binary cubic $x^3 + y^3$ is xy, where $xy = 0$ is the equation of the tangents at the node. We obtain as the only linear functions of U, $\tilde{H}U$ which are decomposable, $x^3 + y^3$ and xyz, the equation $xyz = 0$ gives $xy = 0$ which belongs to the tangents at the node or else $z = 0$, which is the equation of the line through the three inflexions: this line is so obtained a little further on in the text.